Release Notes xxxx-xx-xx : : not ready at all
The AD literature seems to go from miracle to failed miracle almost daily with little attempt to integrate new results with old observations. There are now a lot of data poitns to put together with cause and effect like any TV detective would do or as described in fables like the blind men and the elephant even though apparently Natalie Merchant set it to music. This work tries to fix that and finds that all these things are logically explained by well known issues such as age related digestive defects. Suggestions are further supported by in-house work on optimization of dog diets demonstrating apparent safety and utility of out of favor vitamins that could be impacted by changes in stomach chemistry. Another problem may be politics. One issue may be related to tyrosine metabolism and increased relative abundance of analogs capable of becoming toxic. This immediately jumps out as a contributor to race based AD differences not just social aspects that are touted as they are politically correct. To be sure, constituitive tyrosine metabolizers may have adaptive responses which support this flux with no net "bad" effects elsewhere and theere may turn out not to be any anwyay. In any case, its an important issue and like any other law of nature will not yield to human desire and the only way for people to benefit is follow the data. If you want to fool society, appease angry people, instead of solving problems and eliminating run on sentences for everyone turn back now to your safe space...
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Alzheimer's Disease : 10k maniacs agree its an elephant

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(Dated: March 18, 2026)

Effective treatments for Alzheimer's Disease(AD) have been difficult to create. Much effort has been directed at amyloid beta removal with minimal clinical benefits demonstrated. Several alternative approaches have become popular including the infection hypothesis. Since so many pathogens appear to causes AD an age related vulnerability may be a better focus. Over various time spans, a series of headlines has emerged claiming benefits for various nutrients or supplements such as thiamine, choline, lithium, or NAD+. However, none appear to generate robust recovery. Taking these results and other observations together, similar to the fable of the blind men grasping different part of an elephant, a plausible unifying theme is nutrient deficiency due to age related digestive defects. Isolated profound nutrient deficiencies will not likely exist so obvious attribution of disease to single molecular entity is not a good starting point. Note that this goes beyond failure to absorb nutrients but also the creation of age related toxins, possbly methanol for example, that may need to be mitigated. Adaptive and malapative responses may make such an initial cause difficult to discern. The literature on the gut-brain axis addresses some of these issues but does not try to identify specific causes for GI distrubrances. This work motivates "meal engineering" as an intervention strategy that may be able to correct many of these problems. If this analysis accurately captures cause and effect in the real disease, correction or prevention may be fairly easy with well known small nutrient molecules or addition of acid or chloride to meals. Individual responses to nutrient details will appear to make each case different but still possibly treatable with similar simple remedies. Indeed, there has been some limited success with dietary interventions but these fail to be designed around likely age related causes although they may coincidentally mitigate some of these problems. It may not be possible get all required nutrients from food once damage to digestive system has occured and supplements or other entities may be required. Cations such as metals, protein bound nutrients, amino acids, and lipophilic nutrients would likely be the first suspects. In particular, its likely that vitamin K and copper, along with more accepted components such as amino acids and SMVT substrates all need to be included to replace the younger absorption profile. However, brute force supplementation without regard to formulation may not work due to chemical and metabolic interactions. Two important amino acids with solubility issues in particular, tyrosine and tryptophan, and critical but reactive metals such as copper may require special attention. There are also possible political concerns if there is a hesitancy to link tyrosine metabolism to dementia. However, as always, social influences will not change natural phenomena.

CONTENTS

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I. INTRODUCTION

Alzheimer's disease [44] remains an unresolved problem with important impact to society [37] despite all the money and effort directed at it for the last several decades [51] . Anti-amyloid products have been the major focus but continue to show questionable net clinical benefiti [190] [170] and do not always get FDA approval even with creative trial design [67] . This is in spite of the fact that as early as 2002 some had noted too many inconsistent observations about amyloid or tau for them to be serious targets [162] . Several other approaches and theories have been developed with various levels of support although it would be difficult to say that a specific next-best has emerged. The thesis of this work is that all these distinct approaches can be resolved by looking upstream for initial causes for these diverse observations about the disease. Before detailing these "pieces of the proverbial elephant", some indication of the problem may be appreciated with recent effort to emply AI against AD. Given the different aspects of AD, often described as "multifactorial" or "heterogenous" emerging AI has been proposed to lead a response [96]. [72]. This may yield results, its progressing too rapidly to rule out, but often it is limited by traning not on raw data but human commentary and a statistical rather than "model of reality" based view. Others are using AI for precision medicine thinking that patient heterogeneity is a problem [225]. Some are searching for connections between apparently unrelated observations [83] to find tractable intervention strategies similar to the present work. Headlines sometimes appear about new scientific or anecdotal evidence pointing to a new treatment but rarely do these translate into useful therapies. Part of the thesis of this work is that many of these results are useful but tend to be interpretted in isolation and without regard for cause and effect in real disease despite often intricate and detailed biochemical anlyses related to each piece. While some pathway details have been described, the search backward to an "initial" cause is not common. Often the actual cause is treated as irrelevant. "Animal models" may mimic some features of real disease but are predicated on a cause unlikely to be relevant to the real disease [177] and their limitation are becoming more appreciated . By coincidence, sometimes however results may be transferable but don't seem to add to understanding. These will be discussed in more detail in the context of each piece.

The following subsections list some AD related theories or data that motivate the fuller thesis of this work identifying a more of less tractable point of failure which leads to diverse symptoms but yet may be easy to treat. In essence, this work suggests that the best intervention is in the GI tract, similar to dysbiosis theories and malnutrition and diet theories, but with the goal of addressing specific causes of these consequent conditions.

A quick note on taxonomy and focus may be helpful. Dementia in general and AD in paricular are not well understood but can be defined by a few concepts. AD is generally defined by amyloid and tau markers [106] but as mentioned later there is reason to believe these have various reasons to occur. Typically AD is divided into a sporadic and familial or late-onset (LOAD) and early-onset (EOAD) type with work continuuing to explore differences [181] [27] [226] . This work is generally oriented towards late onset sporadic AD although its likely to address many age related issues too as nutrient effects are not entirely brain specific. Indeed sarcopenia will be noted as a correlate or age and disease. The implicit feature of AD assumed through out is that it is a disease of older age and theories need to explain that age dependence. The remaining introduction sections discuss classes of observations or data, "pieces of the elephant" , that the present work claims to organize into diverse manifestations of age related GI impairment as detailed later.

I.1. Some stumbling blocks

Thinking outloud
on the one hand, this doesn't fully capture my disguest but its also too distracting as written ROFL

The hypothesis presented here largely builds on prior work rather than contradicting many of the details but there are a few exceptions to that. When proposing a new theory that is inconsistent with current thinking, and prior work is stalled, it may be that theire are fundamental fallacies in the existing accepted analysis or data. To make the new theory credible it may be important to explicitly identify hidden fallacies or errors in the status quo. Some common ones were enumerated in prior work [143] I listed several common reasons for dead end thinking as the literature appeared to suffer from many. Here too many of those apply and the fixation on amyliud beta, treating numbers to be more normal, ignores upstream cause and effect. AD however is also apparently restricted by social factors as intelligence and behavioral issues may elicit more rage than reason from sponsors. Comforting assertions and assumptions, sometimes to gain more popularity, may be held well beyond what is justified by any data. In other cases, cause and effect may be related in implausible ways which corrupts clear thinking. For example, a recent work looking at sex and race as factors in AD suggests that " educational level" a modifiable risk factor [165]. That may be true to some extent but realistically some may have preexisting brain disfunction making it esssentially impossible to achieve meaningful higher education. And we are already seeing unfortunate consequences of thinking a degress itself, without real accomplishment means anything with the distroted job market [61] [69]. Or, maybe all we need to do is give AD patients a PhD in physics.

I.2. Accumulated Damage Theories

One general class of competing ideas involves "accumulated damage" theories where some cellular components develop non-specific irreprable damage leading to the decline of the brain and body. These can eqaute chronological age with "wearing out" or invoke something like senescence [16] and just accept "it happens." These may concentrate on vasculature and boarder on hopelessness [11] or ROS damaged proteins [135]. Accumulated genetic damage [241] could include a related group of theories. It may include epigentic changes [146] , somatic mutations and mosaicism [127] , and telomere shortening [118] . This work addresses the protein quality issues directly and epigenetic changes tend to be programmed responses to things like nutritional status so they are expected to improve too. One work concerned with epigenetic factors recommended supplements of things like B vitamins and s-adenosyl-L-methionine [189]. Telomere length is not directly considered although it correlates with age some associations with AD are expected and a causal role is being explored [49] . Some correlations between length and cognitive decline appear backwards [100]. A large number of nutrients have various correlations with telomere length such as the usual B-vitamins [172] as well as diets [202] such as the Mediterranean [73]. There also appear to be associations with vitamin K intake [52].

Genetic mosaicism is a normal part of the brain as mutations accumulate from conception [21]. Many of the pathological mutations linked to AD are thought to be encouraged by well known AD risk factors [101] suggesting they can be reduced by addressing these known factors. Stabilization or removal of deleterious somatic mutations may be less obvious. Certainly generic defects in DNA repair may not be remediated solely by diet but defect production and culling of mutants may be possible.

I.3. Genetic Causes

Ultimately all disease is genetic since perfect genes would anticipate and deal with everything that can occur but realistically these consider less common mutations that differ from the larger population. The genetic damage theories have some appeal because several gene variants have been associated with AD and appear in pathways that are plausibly causal although this may create confirmation bias in "relevant" gene selection. One recent work notes hundreds of genes have been implicated and points out that amyloid beta could be a response to stimuli such as, " including infectious agents (herpes simplex, chlamydia pneumonia, and Borrelia burgdorferi) as well as Vitamin A deficiency, hypercholesterolaemia, hyperhomocysteinaemia or folate deficiency, oestrogen depletion, cerebral nerve growth factor (NGF) deprivation, diabetes, cerebral hypoperfusion (leading to hypoxia and hypoglycaemia) " [34]. Then the above concentrates on immune related genes. As "genes" are nominally constant through out life, their roles in age related disease may be indirect such as through reduced DNA repair abilities allowing for damage to accumuale faster with age. For the present theory, its worth noting that no popular GI related gene variants have emerged.

Genetics have been widely explored with several common mutations linked to either familial AD(PSEN1,PSEN2, APP) or sporadic late onset (APOE) disease but with many others possibly involved [12] . Among these are things like SLC10A2 and a variety of lipid, calcium, or immune related genes. APOE is associated with many vitamins including vitamin K. APOE was identified as important for vitamin K transport to bones as early as 1996 with apoe4 noted for association with lower serum K and increased fracture risk [116] and it was noted response to supplementation occurs over months. However, a later work found "superior vitamin K status is associated with the apoE4 genotype in healthy older individuals from China and the UK" [231] .

Thinking outloud
find the work on secretase and hair color for example

A some human mutations identified in specific populations, along with humanized genes, have been used to create animal models of AD. The 5xFAD mouse overexpresses mutations in APP and MSEN1 and mimics some features of real disease although not NFT's [167]. EFAD mice include a human APOE gene [212] but still interpretation of any intervention has to be tempered with an understanding of the assumptions inherent in these models. Interestingly, the 5xFAD mice respond differently to diet than wild type. One study found some apparent decoupling of results,

> Thus, 5XFAD mice exhibited greater susceptibility to highfat diet than wildtype mice regarding cerebrovascular injury and cognitive impairment. On the other hand, 5XFAD mice fed highfat diet exhibited much less increase in body weight, white adipose tissue weight, and adipocyte size than their wildtype counterparts. Highfat diet significantly impaired glucose tolerance in wildtype mice but not in 5XFAD

> mice. Thus, 5XFAD mice had much less susceptibility to highfatdietinduced metabolic disorders than wildtype mice. [131].

These models were largely designed with the assumption that accumulation of amyloid beta was the central feature to replicate but that may turn out to not be correct. Since the cause and effect in real disease would only be captured by luck there could be a lot of limitations from any of these models. Interestingly, transgenic mice may show intestinal symptoms correlating with brain pathology [139].

AD risk varies significantly by sex and race [130] making the related genes suspects. However, other factors such as sociocultural [68] "unique social and environmental pressures" [158] are often blamed. Correlates of sex such as estrogen production may be a more direct factor [185].

Some polymorphisms of the digestive tract are thought to significantly alter drug disposition and they can be race related [75].

More generally, there could be important race or sex based differences in many catagories that are due to easy to treat malnutrition and a focus on social factors distracts from a real solution. Vitamin A related polymorphisms were found to correlate with ethnic group of vitamin A status in a sample of pregnant women [210]. One study of DIO2 polymorphisms and AD among various groups demonstrated a possible contributor to AD in blacks along with a lot of transcription data [152] that may be a good starting point for further hypotheses.

In fact, more directly relevant genes have been found in the black US population. A dopamine receptor D2 variant (DRD2 Taq1AA1) more common in blacks than whites likely associates with AD risk [24]. Presumably variants would differ in some binding kinetics including to dopamin analogs that could become important as parent neurotransmitter becomes more scarce with an aging difective tract.

I.4. Clues from Transcriptional Associations

Besdies mutation data, transcriptome analysis has found genes which are over or under transcribed in Alzheimer's Disease. As with most data classes, this has not created an unambiguous path to an intervention and cause and effect, good or bad, remains unresolved for most of these genes and their products. A gene may be over or undertranscribed depending on how it is regulated. Functional feedback could signal for more of a dysfunctional product to be created or one that can not function for some system related reason. So, typical interpretations may be missing important bpoints that may fit well into the model proposed herein.

A 2025 article found ADAMTS2 to be one of the genes robustly increased in black and European Alzheimer's brains [216]. Essentially this translates into pervasive "weak structural proteins" as described below. Mice with non-functional ADAMTS-2 develop fragile skin [126]. It is over expressed in mice and humans with hypertrophic hearts and this condition is worse if ADAMTS-2 is lacking [223] The information so far suggests it is upregulated in cases where is may be beneficial but is unable to accomplish whatever the feedback is regulating. In the case of structural proteins, something may be signalling " more or better " but another factor is lacking (ie malnutrition).

A 2021 work described a liver-brain axis and implicated junk food consumption, perhaps decomposed into both nutrient and toxin exposure, as a driver of AD with related changes in expression of genes such as, " TUBB5-HYOU1-SDF2L1-DECR1- CDH1-EGFR-SKP2-SOD1-IRAK1-FOXO1" and corresdpondingly, " [gene signature in the] decline of concurrent liver and brain health. Dysregulated levels of these genes are linked to molecular processes like cellular senescence, hypoxia, glutathione synthesis, amino acid modification, increased nitrogen content, synthesis of BCAAs, cholesterol biosynthesis, steroid hormone signalling and VEGF pathway." [93].

I.5. Infection Hypothesis including Cumulative Damage

Another approach gaining interest is the infectious hypothesis of AD [84] [28] [153] [31] . The basic theory is that one or more pathogens cause AD although many are very common such as P gingivalis [80] , varicella zoster [30] , COVID-19 [15] , the Lyme pathogen [84], and Chlamydia pneumoniae [36]. With one abstract listing several diverse contributors [175],

> viruses (e.g., Herpes simplex type 1, 2, 6A/B; human cytomegalovirus, Epstein-Barr virus, hepatitis C virus, influenza virus, and severe acute respiratory syndrome coronavirus 2, SARS-CoV-2), bacteria (e.g., Treponema pallidum, Borrelia burgdorferi, Chlamydia pneumoniae, Porphyromonas gingivalis, Prevotella intermedia, Tannerella forsythia, Fusobacterium nucleatum, Aggregatibacter actinomycetemcmitans, Eikenella corrodens, Treponema denticola, and Helicobacter pylori), as well as eukaryotic unicellular parasites (e.g., Toxoplasma gondii) may factor into cognitive decline within the context of AD.

Part of the motivation is the lack of a clear consistent pathology related to amyloid beta but an apparent role in the brain's immune response [91] (which also lists many organisms and advocates infection control) [74]. It has also been observed that H pylori infection associates with prevelance of AD [171], an observation considered by the present work to indicate a related vulnerability is causal to both. Neurosyphilis can be difficult to distinguish from AD [156] creating precedent for at least and AD mimic.

In further support of this , a wide range of vaccines has been found to reduce risk of AD or other dementia [137]. Another study found benefits of vaccination and also anti viral treatment for herpes zoster [145]. There is also some indication AD may be transmissible similar to prion diseases [196].

Thinking outloud
microcyline and stroke could be stopping hypoxic virulence from unknown pathogen

The observed age dependence of AD compared to life long exposure to pathogens has to be reconciled with something like a cumulative damage scenario or an age-dependent host response. Many pathogens cause known GI damage that is often ignored. Covid-19 GI damage and interaction with ACE2 receptors has been studied [147] and secondary bacterial infection with gut organisms is common [19]. Varicella zoster similarly can cause GI issues [161].

Some organisms trigger an interferon response leading to the loss of nutrients like tryptophan [237]. Nutritional immunity can be invoked [163] towards various nutrients with particular emphasis on some minerals like copper [42] that may redistribute in ways that become pathological if deficient. While amyloid may have prion like properties, this kind of transmission does not seem likely in the vast majority of cases. Hypermetabolism has been described in Covid 19 although perhaps underestimated [29] .

I.6. Gut-Brain Axis Hypotheses

Another general group of hypotheses involve the gut-brain axis [94] [97] [125] [234] [174] . Typically this points to something such as the microbiome that gets "out of balance" and needs to be adjusted with innoculation or some feedstock mix or fecal transplant. Often nutrition is considered and a detailed analysis of cause and effect between gut and brain is performed. This may lead to similar nutrients as those consisdered here but doesn't usually get to the initial cause hypothesized here nor design a solution around that cause.

I.7. Metabolic or Nutrient Hypotheses

This work will consider many combinations of nutrients that have previously been identified as possibly useful in AD. However, these prior works generally don't try to determine a cause or a reason to single out one or a few nutrients.

Works cataloging the utility of vitamins or derivatives for AD exist [219] [199] but generally treat these things as "one at a time" approaches. Just this year a work suggests combining the usual suspects A,Bx,D,E,K [160] apparently neglecting protein and minerals and without real rationale.

Although at least one review from 2022 recognized the likely need for multiple components [5]. Some mixes such as a mitrochondria nutrient mix [134] have been considered. Often mixes seem to be predicated on old ideas such as a combination of cartenoids, omega-3 fatty acids, and vitamin E which also may demonstrate a passing benefit [166]. The point of this work is to determine, or at least guess based on current knowledge and observations, a better cause to focus the response. Another work combined A,D,E,K and multiple B vitamins as well as probiotics with motivation being oxidative stress, inflammation, and dysbiosis [209]. However, its not clear what is causing all those things or why they matter. In particular, oxidative stress may be a signal for among other things copper deficient mitochondria. Too many to mention employ some combination including obscure or exotic herbs or ingredients [107] often for glib reasons. Some success has been observed in probiotic trials [173] which of course likely effect GI chemsitry. Dietary patterns, similar to issues with "meals" discussed here, have also gotten some attention for empirical correlation with dementia such as a MIND pattern [47], Medietteranean and DASH [224], Japanese [239], and there are others [200] and other outcomes considered. A high sugar dietary pattern was analyzed with HR as a function of sugar intake showing an optimum but increasing beyond that [240]. This will be useful later if considered in the context of auto-brewery syndrome and methanol exposure. The partially beneficial ones of course suggest ingredients but not all componets are likely to be beneficial.

I.8. Various Toxin Hypotheses

A variety of theories exist about cumulative exposure to environmental toxins such as "harmful metals, pesticides, agrochemicals, and air pollution " [56] or " heavy metals, flame retardants, and pesticide residue" [77]. Agent orange may produce some similar symptoms as AD [54]. This work will look mostly at toxics thought to be produced endogenously in the impaired stomach. The most prominent of these is methanol and its potential metabolites such as formaldehyde. Methanol has been well explored for cognitive impairment [124] as has formaledhyde [232]. These will be discussed in more detail as the current hypothesis is elaborated. However, a huge array of induced secondary metabolites are possible from any GI tract microbiome so its difficult to rule out anything. Some potentially important ones have been discussed [242]. Conversely, some marine dinoflagellates toxins have been considered for reduction in amyloid burden [26]. And its also worth mentioning that acetylcholinesterase inhbitors while usually considered toxic may be therapeutic [204].

II. THE PRESENT HYPOTHESIS

The thesis of this work is similar to some aspects of the gut based hypotheses except it directs attention to the likely underlying cause of dysbiosis or other phenomena linked to gut-brain-axis theories. Some works recognize an association between AD GI tract disorders but then tend to suggest things like," antibiotic therapy, probiotics and prebiotics, and faecal transplantation are being considered as possible therapeutic options" [39] as in the dysbiosis theories.

A 2025 work did elaborate on the concept of pre-digested food for special populations such as the elderly with a detailed review of some issues [227] and it discusses some approaches such as fermentation that may be explored herein. However, it does not appear to address toxin production by the distressed GI tract.

As no particularly good suspect has emerged, it may be worth looking at all the data and trying to guess at age related vulnerabilities that cause infections to lead to dementia, digestive disturbances, and classic symptoms such as amyloid build ups. Nutritional defects make some sense as frailty, more than age per se, may be a correlate of age related diseases.

II.1. brain microbiome

In a prior work [142] I interpreted a brain microbiome result to describe some host factors that would explain differences in the most and least abundant organisms in AD and healthy brains post mortem. Microbiome studies on low-biomass samples have been criticized previously for susceptibility to various errors [113]. While the existence of a brain microbiome is controversial due to the expected low absolute abudance and opportunity for contamination, the analysis is somewhat robust in that contamination from other parts of the patient should still reflect differential body nutrient levels. One argument against contamination was similarity to an unrelated work on uterine microbiome and possible symbiotic role of these organisms but still a lot of coincidences can occur. This work generally pointed to deficiencies in lipophilic amino acids but also suggested additional methanol in the AD patients. One likely source for this is the GI tract due to changes in microbiome or initially changes in chemistry. Other toxins could associate with methanol production too and a good strategy may be to restore GI chemistry back to youthful state.

II.2. methanol and auto brewery syndrome

Methanol can be generated in the body from ethanol [57] while both ethanol and methanol circulate in a normal person at the .5-1.5mg/L level and may be produced from carbohydrate consumption [98]. Endogenous ethanol production in the stomach has been observed [208] and considered. One post mortem found high ethanol in the stomach attributed to Candida glabrata [120]. A recent review of auto-brewery syndome suggested specific organisms may not be the issue as much as "...the use of medications, notably proton pump inhibitors and antibiotics. When combined with a Western diet rich in fat and simple sugars, these factors significantly contribute to altering the gut environment and subsequently, ethanol production." [155] although a list of ethanol producers has been created [151]. In one patient an oral glucose tolerance test achieved an ethanol concentration of 315mg/dL in the blood [198].

II.3. benzoate

Sodium benzoate has been explored as a therapy for dementia apparently due to D-amino acid oxidase inhibition [133] and may have a shown a benefit for women in early phase AD and also increased some hormone levels [132]. A more recent review suggested it may have an effect in various mental conditions that is likely stronger in women than men [128]. Another review cautiously suggested a benefit in early stage AD but noted the small sample size and reliance on small number of trials [140]. It may only function in cases where dementia would otherwise be dominated by methanol or some other fermentation related products. Studies have been conducted on sodium benzoate and the GI microbiome largely due to concerns about toxicity as a food additive and results have been mixed although apparently well established for benefits in farm animals [123].

Thinking outloud
move some of this to the specifics sections later

Amino acids are thought to play a role in AD but several results suggest them to be unimportant largely probably due to glib or superficial interpretations.

II.4. tryptophan

Tryptophan is a good suspect for many reasons but various supplementation trials have yet to show an interesting benfit in AD. In this work, the failures will be attributed to formulation, a need for other componets of the limiting pathways, and failure to address any sources of toxins like methanol from the pathological GI chemistry which may exist.

Among treated metabolic syndrome patients, the Kynurenine to tryptophan ratio, as well as inverse C20:3 and lysine amounts, correlates with the risk of mild cognitive impairment [95]. This work also mentions patient heterogeneity which may be an important factor in testing nutritional interventions.

Tryptophan has as a contributor to many diseases has been reviewed in several publications [176] [102].

A mixture of 9 essential amino acids includinc tryptophan given for 28 days failed to show any meaninful benefits [213] but this may be toolittle, too short a time, the formulation was not clear, and given together they can compete for entry into CNS. Other factors may be needed to get uptake and transport. A tryptophan dipeptide, a product of fermentation which may have improved absorption, did have some effects in mice [13]. At 25mg/kg BW for 12 weeks, elderly with mood disorders showed some improvements [43]/ although formulation was not clear. A variety of tryptophan derivatives have been explored such as 5-HTP [3] and indole 3-propionic acid [117] have also been explored.

Once nutrient issues occur, they can cause other problems such as transport reduction making simple supplementation less likely to work easily. One small study in rats found an age dependence of transport across the BBB both saturable and passive diffusion rates decreased in older animals [214]. This would suggest that at a constant blood level less enters the brain and supplementation with trytpohan may be futile. However, the transport restriction may be simply due to loss of other nutrients with age. Of particular interest was the 1991 observation that oleic acid can open the blood brain barrier [211] which has been observed to change with age and AD [187]. The quality of the BBB could then be responsible for many nutrient deficiencies in the brain but as with other structural components may be limited by nutrient uptake from the GI tract. Note too these issues are not mutually exclusive. Tryptohan occurs as a conserved amino acid in transporters and at membrane interfaces. Once tryptophan is depleted transport and barrier may suffer.

II.5. histidine

Another boarderline amino acid may be histidine which is precursor to histidine. One study in mice claimed it improves response to chronic hypoperfusion [206] although data suggest small differences in quantifies explored. A study a few decades ago did find reduced regional histidine in post mortem AD brains [150]. At least one clinical trial in AD was considered, NCT06169826 `https://clinicaltrials.gov/study/NCT06169826` but apparently withdrawn.

It may provide protection against ischemia [129] and reduced oxygenation may be a factor in aging.

II.6. arginine

Oral arginine as recently described as suppressing amyloid pathology in animal models of AD [71].

.

II.7. glp-1

GLP-1's have finally been getting popular press attention due to related nutrient deficiencies such as scurvy [203]. In at least one case report, with no suggested GLP-1 usage, severe GI symptoms may occur from reduced retroperitoneal fat pad buffering the aorta and superior mesentaeric artery combined with other problems like scurvy [8].

And indeed scury, another state creating "weak structural proteins"

Thinking outloud
my new catch phrase, probably including the collagen sca

has been noted in the literature in particular with colonic pseudoobstruction [59]. Although there has also been concern raised about under estimating the harm of malnutrition related to their use [64]. Other nutrients of concern include vitamin D, iron, calcium, protein, B-1 and B-12 [218]

Particular concerns have been raised about the hidden danger of lean muscle mass loss and resulting lack of resilience. [217]. The GLP-1 data may also motivate an interest. GLP-1's apparently are causing non-specific nutrient deficiencies maybe similar to problems common in old age. Recent clinical trials showed either no benefit in AD with the Novo Norodisk product [2]

Thinking outloud
does the author realize running against "other" antidiabetic drugs is misleading? ROFL

or a positive trial [62] appeared to be cherry picked. A Novo-Nordisk presentation on the results, `https://sciencehub.novonordisk.com/content/dam/sciencehub/global/en/congresses-and-scientific-publications/congresses/ctad2025/johannsen1/sliders/Symposium_4_Cummings_Jeffrey_V2_OH.pdf` does suggest largely no effect either good or bad although some of the side effects may be unpleasant. Its likely that "early stage AD" implies existence of important nutrient limitations have already been there for years so the differential impact, despite GI symptoms, may be minor.

A pooled analysis of 4 trials and 112 AD and MCI patients showed no significant outcomes differences and numerical harm well below significance [168].

The emergence of clinically important aged appearance (" Ozempic face") [164] [194] and concerns about neuriological adverse events [41] or "brain fog"i anecdotes [215] are well known despite several works suggesting they should be brain beneficial

. There are a lot of lessons here but the immediate thing of relevance is that nutrient limitation that is can be "plausibly denied" appears to generate some symptoms of aging and more to the point frailty. Interestingly, at least one study found a correlation between perceived face age and dementia risk [230].

Evidence pointing towards a cognitive benefit for GLP-1's may be a bit misleading. In one case, they were run against sulfonylureas which have known relevant side effects. In other cases, words suggesting a cognitive benefit in actuality mean that some risk factor has been reduced without clear evidence of real clinical benefit against dementia.

II.8. race and sex

Women and blacks appear to be at higher risk of AD than European males. Factors unrelated to the disease per se may confound results but that is the case with even politically correct association studies. A small 1996 study found that blacks has a higher found higher pH and HCO3 secretion as well as higher prevelance of Helicobacter pylori infection and gastritis [50]. Difference in "normal" protein digestion in healthy males and females appear to be known and have been considered already for a basis of "engineering foods for women" [121]. Note that this empirical difference does not rely on a specific reason although the chemistry may be informative for meal design. A small 1982 study of relatives found that older and female probands had lower acid output although some results depend on how the output is normalized [112]. One study comparing healthy 23-42 year olds with 44-71 found more acid secretion in the older group [78].

While one study did find differences in serum tyrosine levels between older people with and without AD [136] blood levels may vary due to transport issues.

One obvious issue with skin pigment is tryosine metabolism. Both the depltion of tyrosine and generation of toxic analogs could be considered. It is possible that constituitive skin pigmentation (CSP) evolved with adaptive mechanisms that may be instructive to investigate. CSP evolution is usually described as an issue of sun exposure with damage and vitamin D synthesis being driving forces in tone [92].

While CSP may predispose to various maladies, it may also be easily treated if existing adpatations are not already active. Indeed, speculation on beneficial tyrosine analogs can not be dismissed.

Thinking outloud
goog ai went wild with "dementia skin pigmenta

And melanin itself is not inert once formed especially in the presence of UV light [207]. A link between dementia and melanin has been considered by many authors. A study of several thousand Taiwanese demonstrated a hazard ratio of over 12 for AD in vitiligo sufferers versus control (HR of 5.3 for any dementia) [38]. Vitiligo is an autoimmune disease making interpretation difficult but it does suggest an age related problem with melanin due to covert pathogen or defective products.

Fringe ideas in the mainstream litearture include ability of melanin to transfer light energy to the cell for ATP creation and loass impairs function [18] . More plausibly, loss of something like tyrosine or copper limits pigment synthesis in some idiosyncratic manner or more to the point consituitive pigmentation requires resources that are no longer plentiful.

In zebrafish, there was an unexpected link between PSEN2 and pigmentation [81] of unknown significance.

Tyrosine anecdotes around menopause may be one indicator. While not considered essential as it can be derived from phenyalanine, it is possible that overall supply of the precursor and regulatory systems reduce tyrosine production pathologically. This is most likely during times of "stress" and shortages. Interestingly in connection with GLP-1 there is a tyrosine dipeptide apparently acting as a signalling molecule Amino acids such as tyrosine may have limited availability in cell cultures and dipeptides have been explored [104] for many years now. Also of interest is the need to supplement tyrosine as a dipeptide to Chinese hamseter ovary cells in bioreactors [222] with work showing possible pickiness of cells for uptake although translation to human GI tract remains to be done. Similar problems have been noted for tyrosine usage in humans from either IV injection or parenteral feeding [138]. Tyrosine-tyrosine peptide is considered a hormone and may be responsible for anorexia in urea cycle disorder patients [159].

Comparison to other racial groups does not appear informative as data are all complicated.

Some context for phenylanine to tyrosine conversion as part of the larger protein turnover dynamics has been explored [76].

Indeed GLP-1 is often considered with PYY for understanding obesity signalling [55]. While its unclear if PYY can replace the GLP-1 drugs, if supplementation is attempted it will benefit a lot from recognizing the issues explored here. Dipeptides or reproduction of "best' GI chemistry may be more important than simply adding tyrosine to diet.

A variety of nutrient deficiencies have been observed to correlate with AD risk such as B,D, and E [45].

II.9. Copper

Copper has a lot of motivation behind it but its rich chemistry may make it one of the more difficult to study. Copper is difficult to characterize but some results may motivate its importance. Overall it appears that AD brains are copper deficient and well formulated supplementation may be beneficial but it can not be excluded yet that other nutrients are limiting its distribution. At least one case report exists of a defect in ceruloplasmin resulting in chorea and dementia with iron deposition that resolved somewhat with fresh frozen plasma [235]. As early as 2016, copper overload was suggested for regenerative medicine and boosting mitochondria performance[191].

Sometimes there is a tendency to think of local levels as a passive result of supply but often complicated signalling generates overall results. As early as 2009, the paradox of excess copper with amyloid beta coincident with tissue copper deficiency was recognized and explained to some degree [89]. In prior work with dogs, the inability of chelation to produce clinical benefit in animals with hepatic copper accumulation pointed to a bottleneck in transport rather than a global excess[143]. Also in the above work it was noted that "copper toxicity" varied widely with the background diet suggesting it has to do with chemical reactions in the meal. Rats raised on a copper deficient diet showed increases in adrenal dopamine-beta-monooxygenase mRna, protein levels, and enzyme activity in decreasing order [180]. This suggests that feedback mechanisms are attempting to increase activity levels with greater transcription rates but unable to metallize the extra translation products. A 2017 mass spectrometry study on post mortem brains found overall decreased copper in AD brains similar to that in Menkes' Disease [229].

One interesting in vitro work was able to image intracellular Cu resolving oxidation states [201]. The work suggests amyloid beta can act similar to a siderophore and import copper, instead of iron, into the cell leading to death unless rescued with a chelator. The authors note the probe may perturb physiology and that Cu is released into the synaptic cleft. Taken together with existing literature, this tends to point to a bottleneck that may be partially neuron specific. Eliminating the excess copper may prevent death in vitro but may not restore normal brain function.

II.10. Vitamin K

For whatever reason, vitamin K is often ignored even though interactions with APOE and appearance in "healthy" foods motivates at least an association with health.

An early proposal for vitamin K deficiency was described in 2000 citing "regulation of sulfotransferase activity and the activity of a growth factor/tyrosine kinase receptor (Gas 6/Axl) " [6].

In the last few years many works have explored vitamin K in old age conditions [63] [103] including association between vitamin K intake and cognition [25]. As bleeding or microbleeding is an issue in AD, its worth noting that 82 percent of "chronic stroke subjects" were vitamin K intake deficient and excessive vitamin E may also be a common problem [79]. The above work also describes more direct effects of vitamin K in AD including interactions with amyloid beta. (there are works that concentrate on emerging problems with vitamin E supplementation too, for example [111])

II.11. NAD+

NAD+ deficiency has also been explored in AD [88] and other age related degenerative conditions [122]. Disturbances of NAD+ can influence trajectory of viruses such as Japanese Encephalitis Virus in mice and human cells [108].

II.12. Choline

Choline has been the topic of several recent works [53] [221] adding to a long history for it [220] [40] and related pathways [70] including cholinesterase inhibitors [20]. The 3xTg-AD mouse apparently has low levels of circulating choline [99] and that may be instructive to determine if the model genetics relate to this. While often related to acetylcholine, its important to remember its role in nutrient absorption.

II.13. Taurine

Taurine has been proposed as possible disease modifying supplement [87].

II.14. SMVT Substrates

Biotin and the SMVT substrates [195] have also come up in AD work and dementia more generally [197] [46] . Biotin is critical for many processes and can be obtained from digestion or intestinal microbe production [105]. A 2026 study of dietary biotin and AD risk combined with neuropathology results in biotin deficient mice suggested a biotin decieicny as a contributing factor to AD [238].

II.15. Thiamine

Thiamine is also being considered [65] and that too is through to have acid sensitive absorption [119]. A 2026 work identified a new mechanism by which it could mitigate problems with amyloid beta by suppressing HIF-1a [10]. A 2012 study did show some benefits from supplementation in the elderly but due to poor absorption parental dosing was mentioned [184]. It seems that many studies on isolated nurients have come to similar conclusions but not attempted to think through cause and effect more generally. A 1988 study claimed to use a niacinamide placebo [23] which motivates another problem of an unintentionally active placebo. And in fact niacinamide has been discussed as a treatment too [115].

II.16. Lithium

Lithium has recently been rediscovered [14] although positive review articles date back to at least 2012 [66] or so [148] and its not clear what happened over the intervening decade.

In 1971, it was observed that lithium could impact carbohydrate generally and inositol status in the brain [7]. In another manuscript considering inositol as a dog supplement (unpublished result)

but as of 2005 it could not be determined if lithium or inositol was more directly causal to brain changes that matter in the clinic [82]. This is likely a recurring issue in Alzheimer's and medicine overall. Associations are difficult to map to an actionable cause and effect sequence that can be beneficially modified. It is the main theme of this work that the primary cause is well removed from these associations and the elephant is not apparent from the sum of its parts. It is likely that both inositl and lithium are impacted by some common cause and simply supplementing one or the other or both can make some improvements but is a deadend to a cure.

II.17. Bicarbonate, Chloride, Phosphate

There may be some intersection between detergent chemisty and conditions needed in the GI tract to absorb vitamins and retard toxin impact.

Thinking outloud

III. COMMON AGE-RELATED DIGESTION DEFECTS

To begin to make sense of the above and look towards the design of a better intervention, some knowledge of likely defects and resulting deficiencies needs to be established. Existing literature is a guide but like all published works needs to be considered in light of available data. Most works tend to minimize the importance of typical GI aging on health apparently mostly based on assertions and assumptions. Contradictory observations would need to be considered carefully and its likely intervention design will be somewhat empirical hoping to gain information as work continues. For example Chapter 8 in a 2014 book [109] suggests a large reserve capacity and no real effects absent disease with lowered uptake being a consequence of reduced protein mass. In what follows the "reduced protein mass" may be considered a problem in itself consequent to reduced nutrient uptake for reasons yet to be determined for this work. By 2025 the problem of protein uptake seems to be recognized [85] [183]. Cause and effect may be quite difficult to trace however. Consider the case of aginng colonic muscles cells with reduced performance related to L-type Ca currents [32]. Resting $[Ca^{++}]$ is remarkably low with a large ratio to extracellular levels [110] and therefore sensitive to leaks. In fact as early as 2003 reduced response and intracellular accumulation were considered as a reason for "age-related learning deficits" [149]. Calcium leak may occur for several reasons [141] [144] and mutation related disease suggest problems with high-infildelity translation that could occur with amino acid imbalances.

By 2025, methods to simulate elderly digestion in vitro were discussed because age-related GI properties were thought to be important for health related outcomes in the growing elderly population [228].

Thinking outloud
probably a bad citation, low quality journal doi info missing lol [1]

In extreme cases such as GI resection, problems related to malnutrition such as increased abscess risk and reduced ability to produce collagen for wound healing and other needs [1] . ADAMTS2 upregulation in the case of AD was part of the motivation to create this work as it suggests a chronic problem in collagen quality.

By 1992, common themes in impairment reports were emerging. One review suggested decreased taste and hypochlorhydria with decreased uptake of B-12 and minerals but increased absorption of lipids [192]. A small 1997 study of independently living people over 65 found about 90 percent had basal pH under 3.5 [90] although there is no indication of how this compares to the young or very old.

A study of 79 healthy elderly people found some indications of lower acidity than young people but did not consider it relevant in most individuals at least for drug absorption [193].

The pancreas and intestines may also be effected. One review mentions decreases in calcium, zinc, and magnesium but not copper in old rats [86]. However, the pancreas and intestines must work with whatever the stomach produces with the meal. Earlier, I considered that lack of an early low pH step may produce more idiosyncratic results as some mineral can irreversibly precipitate [143].

Around 2004, it was thought that reserve capacity was sufficient although some reduced absorption of vitamins such as A,D,K, and B6 but tended to minimize significance for treating aging [60]. One problem however is that even today sufficient intake and uptake may not be clear and the impact of multiple "low levels" could invoke adpative responses.

Today however protein deficiency appears to be reasonably well accepted as an issue with aging and its manifestation as sarcopenia [182]. What mey be less apparent is any changes in quality such as due to translational infidelity or lack of "other stuff." Sarcopenia or frailty appear to be highly correlated with vulnerability to several "age related" conditions [233].

Further, the impact of drugs common in the elderly on nutrient status is becoming recognized [17].

Atrophic gastritris is one form of digestive problem. It may be silent, associated with non-acid reflux, and lead directly to deficiencies of B-12 and iron [188].

The list of effected nutrients may be much larger however due to simple lack of data. Vitamin C destruction, rather than simple malabsorption, may be significant with increased stomach pH as well as calcium salt dissolution and absorption rates [35]. Presumably other cations than iron and calcium are impacted with copper being an important candidate likely to be ignored. Some studies suggest elevated serum levels with H pylori positive gastritis [9] which may be expected as Ceruloplasmin is an acute phase protein and blood levels may reflect an immediate stress. Generally malabsorption of copper or excess zinc was described in 2008 as uncommon but increasingly recognized [33].

IV. MEAL ENGINEERING CONSIDERATIONS

Based on the above, the goal then is to replace nutrients and remove toxics that are more common in old age. Return to a youthful GI tract may be a reasonable goal although it may be possible to do even better. Conceivably if nutrient uptake can overcome GI limitations some healing may be observed making possible a return to normal food for a while.

Nutrients need only be supplied over ”relevant” time scales. This can vary a lot from nutrient to nutrient and allows better segregation and rotation than trying to get everything everyday. Incompatibilities include chemistry and competition for receptors or enzymes or transporters etc.

Not every meals needs to be the same and in fact currently with the dogs one meal contains most things while a second is largely reserved for copper.

IV.1. Reduced toxins : chloride and benzoate

IV.2. Solubility : chloride, protons, phosphate, emulsifiers

IV.3. Free nutrients

V. CANDIDATE INGREDIENTS

Candidates are drawn from prior thoughts on food and vitamin supplements along with the presumed GI disturbances common in old age.

Any foods or supplements reputed to improve some age related condition could be examined. In many cases the folklore appears to be a useful starting point as it may rationalize well with the above theory or be supported by in-house work here with dog feeding.

One recent work on pre-digested food for groups suc as the elderly gives list of some ideas [227].

Whey peptides improved some functions in healthy people thought to be due to Trp-Tyr peptides [114].

Fermented foods meet many of the criteria for obtaining nutrients with age impaired digestions. They tend to have low pH, although a group of alkaline fermentation products is known [169] , and can include pH as part of the definition, and essential molecules such as free amino acids and vitamin k. They seem to be associated with reduced dementia risk [178]

Olive oil has been an important part of in-house dog diets (manuscript in preparation) as well as being a highlighted component of the Mediterranean Diet. A variety of possible causal pathways could be imagined for it but here it is considered somewhat uniquely, along with lecithin and phosphorous sources, as an absorption enhancer. Its interesting to note, although of unknown significance, that yeast with a higher amount of membrane oleic acid have higher ethanol tolerance [236]. As part of a meal formulation it may help with uptake of lipophilic nutrients.

Its interesting to note that while choline chloride was being explored under non-physiological conditions (120 C) for delignification of biomass (something similar to digestion of natural foods) chloride was found to be the only important component with NaCl working just as well [205] despite all the interest in deep eutectic solvents etc. Chloride was also thought to be reasonable for copper extraction compared to other biocompatible chemistries [143].

2 related objectives are to improve chloride levels and reduce pH. In the results cited previously, acid came from citric acid and HCl supplement formulations. Formulations that use HCl as a raw component would require great care and may be avoided initially. Some soft drinks such as diet cola beverages contain phosphoric acid that may have some benefits. Chloride was largely from potassium chloride and again added HCl.

Title			
Ingredient	Amount	InCompatible	Remarks
Essential Amino Acids			
Essential Fats			
Essential Metals			
B-vitamins			
Misc vitamins			
Ion control			
Digestion Improvement			
Microbiology/Toxin Control			

TABLE I. Catagories are never clean since small molecules always do many things and duplicates will be included

VI. DEVELOPMENT PATHS

Since many candidate components are GRAS, its possible to outline informal or ”crowd sourced” data from motivated parties such as caregivers. Concerns about malicious data due to various incentives as well as other issues would have to be considered in detail. I have been using MUQED daily to record dog diets and relate to outcomes. This helped to identify olive oil as an improtant component although it may have a long response time if other sources of oleic acid such as chicken are included in the diet. The added workflow just requires a few minutes after each meal or dosing to enter whatever was given and have it checked for basic syntax. I use ”vi” although more sophisticated or voice entry may be easy to do. The current implementation of MUQED creates a small vocabulary of foods, supplements and outcomes as well as a simply syntax for entering amounts right after or even skeletons before consumption. As the current implementation is designed for just ”a few” people or dogs, everything is done with text files locally although collation with a standard vocabulary could be done with better scaling implementations.

My in-house results with dogs does suggest some folk lore or anecdotes may have useful signals but will require more data to sort out and a MUQED-like approach may fill the void.

Even if this fails to mitigate disease, it may not be a frivolous effort especially if any trends are observed with prescription medicines.

VII. BIGGEST PROBLEMS

Thinking outloud
lot of issues here include off-target effects, accurate prescribing patterns, population differences between users and non uers, any real effect etc. AGA seems to be pushing this things probably get ad revenue lol

As proton concentration is thought to be a big issue, it may be surprising that chronic PPI use does not lead to obvious epidemic of AD. Although in older people they may not have much real impact as stomach acid is already likely to be low (if PPI's are effective at anything it is likely through some other mechanism. If you can't tolerate stomach acid you are going to be malnourished anyway - something is wrong so you may expect accurately prescribed PPI users to already be predisposed to dementia). For H pylori control, it turns out they likely prevent a bacterial pump rather than a host pump from functioning leading to reduction [22]. In general they lead to less healthy microbioome and over abundance of Streptococcus that may also produce symptoms in patients [157]. Its worth noting that some species can secrete virulence factors inducible by a return to low pH [58]. The PPI's may appear to relieve symptoms but allow the organisms to proliferate making the situation worse with a return to physiological acid levels. (lol).

Current evidence is conflicting even among older or sick people who may already have GI impairments minimizing any effect of PPI's. A 2023 study of a 1.98 million Danish cohort found an incidence of any dementia ratio between PPI users and non-users of 1.36 but reducing with age [179]. The American Gastroenterological Association appears to advance the position that this is unlikely. One post hoc analysis of the ASPREE aspirin trial involving people older than 65, found no association between PPI use and dementia [154]. Reported results are heavily corrected for various factors and the raw HR was not immediately obvious. It is possible the low dose aspirin has some impact but difficult to tell as presented.

Protons and chloride appear to be differentially regulated and PPI's do not stop normal chloride secretion [48] despite the fact they may be used to prevent excess fecal chloride excretion in abnormal cases [4]. There are several chloride transporters in the stomach including calcium activated [186] that may decouple from PPI's. The cystic fibrosis literature may provide some clues.

However, the theory here is a bit empirical and recognizes other factors than pH alone may be important. With a non-zero supply of most required molecules adaptive responses may be able to move the effects of the deficiencies around making the clinical presentation difficult to trace to original problem.

Even if the theory is sound and eventually is proven useful as a starting point, social issues are a big problem. For whatever reason amyloid drugs appear to be getting a more favorable response than is supported by known data. Anything involving skin pigment will attract angry people.

VIII. CONCLUSIONS

The present work seeks to find a robust intvention against AD, specifically LOAD but more generally most age related conditions, by engineering meals to mitigate or fix GI impairments that can reduce nutrient uptake and increase toxin production. It suggests that the sum of all the partial theories can best be described by this common source of failure or it identified the proverbial elephant from distinct observations of the pieces. One non-specific symptom is likely weak structural proteins as amino acids and various factors will be deficient. While several paths for investigation are possible, informal approaches by motivated parties may help avoid some problems of the past efforts such as clinging to a familiar target instead of following the evolving data and avoiding ill posed questions such as the action of a single molecular entity on a diverse randomized population instead of the components for even a single cure specialized to even one person.

IX. SUPPLEMENTAL INFORMATION

Example of MUQED output on dogs over several years, https://github.com/mmarchywka/dogdata and some MUQED source code https://github.com/mmarchywka/muqed_util

IX.1. Computer Code

X. BIBLIOGRAPHY

[1] Malabsorption syndromes' influence on chronic wound healing in patients with gastrointestinal resection. doi:10.71010/ajcmr.2025-e227.

[2] Popular obesity drug fails in hotly anticipated alzheimer's trials, 11 2025. doi:10.1126/science.zvgswf0.

[3] Hakeem Adekunle and Olalekan Balogun. Tryptophan and htp supplementation in the treatment of cognitive and mood disorders: A systematic review and meta-analysis. *Journal of Gynecological & Obstetrical Research*, 06 2025. doi:10.61440/jgor.2025.v3.33.

[4] Berendt W. Aichbichler, Charles H. Zerr, Carol A. Santa Ana, Jack L. Porter, and John S. Fordtran. Proton-pump inhibition of gastric chloride secretion in congenital chloridorrhea. *New England Journal of Medicine*, 336, 01 1997. doi:10.1056/nejm199701093360205.

[5] Jahangir Alam. Vitamins: a nutritional intervention to modulate the alzheimer's disease progression. *Nutritional Neuroscience*, 25, 05 2022. doi:10.1080/1028415x.2020.1826762.

[6] A.C. Allison. The possible role of vitamin k deficiency in the pathogenesis of alzheimer's disease and in augmenting brain damage associated with cardiovascular disease. *Medical Hypotheses*, 57(2):151–155, 2001. URL: https://www.sciencedirect.com/science/article/pii/S0306987701913076, doi:10.1054/mehy.2001.1307.

[7] J. H. ALLISON and M. A. STEWART. Reduced brain inositol in lithium-treated rats. *Nature New Biology*, 233, 10 1971. URL: https://www.nature.com/articles/newbio233267a0, doi:10.1038/newbio233267a0.

[8] Alaa Alsarhan, Ahmad Hakim Alhamid, Jamal Ataya, Hadi Alabdullah, Mohammad Alhasan Almasri, and Naser Khaled Hydar. Superior mesenteric artery syndrome in a child with scurvy and long-standing growth failure: a case report. *Journal of medical case reports*, page 55, 01 2026. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC12865927/, doi:10.1186/s13256-025-05767-1.

[9] Rana T. Altaee. Determination the levels of some trace elements and electrolytes in patients with h. pylori positive chronic gastritis. *College of Basic Education Research Journal*, 21, 06 2025. doi:10.33899/berj.2025.vol21.iss2.39781.

[10] Yasmin Tarek Anderson, Katherine Priest, and Jason Zastre. Vitamin b1 protects against abeta(1-42)-induced hif-1alpha activation and neurotoxicity. *Neurochemistry international*, page 106130, 02 2026. URL: https://pubmed.ncbi.nlm.nih.gov/41707701/, doi:10.1016/j.neuint.2026.106130.

[11] Marcus J. Andersson and Jonathan Stone. Best medicine for dementia: The life-long defense of the brain. *Journal of Alzheimer's Disease*, 94, 06 2023. doi:10.3233/jad-230429.

[12] Jesús Andrade-Guerrero, Alberto Santiago-Balmaseda, Paola Jeronimo-Aguilar, Isaac Vargas-Rodríguez, Ana Ruth Cadena-Suárez, Carlos Sánchez-Garibay, Glustein Pozo-Molina, Claudia Fabiola Méndez-Catalá, Maria-del-Carmen Cardenas-Aguayo, Sofía Diaz-Cintra, Mar Pacheco-Herrero, José Luna-Muñoz, and Luis O. Soto-Rojas. Alzheimer's disease: An updated overview of its genetics. *International Journal of Molecular Sciences*, 24, 02 2023. doi:10.3390/ijms24043754.

[13] Yasuhisa Ano, Yuka Yoshino, Toshiko Kutsukake, Rena Ohya, Takafumi Fukuda, Kazuyuki Uchida, Akihiko Takashima, and Hiroyuki Nakayama. Tryptophan-related dipeptides in fermented dairy products suppress microglial activation and prevent cognitive decline. *Aging*, pages 2949–2967, May 2019. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC6555451/, doi:10.18632/aging.101909.

[14] Liviu Aron, Zhen Kai Ngian, Chenxi Qiu, Jaejoon Choi, Marianna Liang, Derek M. Drake, Sara E. Hamplova, Ella K. Lacey, Perle Roche, Monlan Yuan, Saba S. Hazaveh, Eunjung A. Lee, David A. Bennett, and Bruce A. Yankner. Lithium deficiency and the onset of alzheimer's disease. *Nature*, 645, 09 2025. URL: https://www.nature.com/articles/s41586-025-09335-x, doi:10.1038/s41586-025-09335-x.

[15] Ancha Baranova, Hongbao Cao, and Fuquan Zhang. Causal effect of covid19 on alzheimer's disease: A mendelian randomization study. *Journal of Medical Virology*, 95, 01 2023. doi:10.1002/jmv.28107.

[16] Q. Behfar, A. Ramirez Zuniga, and Pamela V. Martino-Adami. Aging, senescence, and dementia. *The Journal of Prevention of Alzheimer's Disease*, 9(3):523–531, Jul 2022. doi:10.14283/jpad.2022.42.

[17] Victoria Bell, Ana Rodrigues, Maria Antoniadou, Marios Peponis, Theodoros Varzakas, and Tito Fernandes. An update on drug-nutrient interactions and dental decay in older adults. *Nutrients*, 15, 12 2023. doi:10.3390/nu15234900.

[18] Stacie Z. Berg and Jonathan Berg. Melanin: a unifying theory of disease as exemplified by parkinson's, alzheimer's, and lewy body dementia. *Frontiers in Immunology*, 14, 09 2023. doi:10.3389/fimmu.2023.1228530.

[19] Lucie Bernard-Raichon, Mericien Venzon, Jon Klein, Jordan E. Axelrad, Chenzhen Zhang, Alexis P. Sullivan, Grant A. Hussey, Arnau Casanovas-Massana, Maria G. Noval, Ana M. Valero-Jimenez, Juan Gago, Gregory Putzel, Alejandro Pironti, Evan Wilder, Lorna E. Thorpe, Dan R. Littman, Meike Dittmann, Kenneth A. Stapleford, Bo Shopsin, Victor J. Torres, Albert I. Ko, Akiko Iwasaki, Ken Cadwell, and Jonas Schluter. Gut microbiome dysbiosis in antibiotic-treated covid-19 patients is associated with microbial translocation and bacteremia. *Nature Communications*, 13, 11 2022. URL: https://www.nature.com/articles/s41467-022-33395-6, doi:10.1038/s41467-022-33395-6.

[20] Jacqueline S Birks. Cholinesterase inhibitors for alzheimer's disease. *Cochrane Database of Systematic Reviews*, 2016, 03 2016. doi:10.1002/14651858.cd005593.

[21] Sara Bizzotto and Christopher A. Walsh. Genetic mosaicism in the human brain: from lineage tracing to neuropsychiatric disorders. *Nature Reviews Neuroscience*, 23, 05 2022. URL: https://www.nature.com/articles/s41583-022-00572-x, doi:10.1038/s41583-022-00572-x.

[22] V. Blandón-Arias, A.M. Ospina-Gil, B.E. Salazar-Giraldo, and T.L. Pérez-Cala. Systematic review: Effect of proton pump inhibitors on helicobacter pylori physiology. *Revista de Gastroenterología de México (English Edition)*, 90(3):388–399, 2025. URL: https://www.sciencedirect.com/science/article/pii/S2255534X25000891, doi:10.1016/j.rgmxen.2025.09.003.

[23] John P. Blass. Thiamine and alzheimer's disease. *Archives of Neurology*, 45, 08 1988. doi:10.1001/archneur.1988.00520320019008.

[24] Kenneth Blum, Rajendra D. Badgaiyan, Georgia M. Dunston, David Baron, Edward J. Modestino, Thomas McLaughlin, Bruce Steinberg, Mark S. Gold, and Marjorie C. Gondré-Lewis. The drd2 taq1a a1 allele may magnify the risk of alzheimer's in aging african-americans. *Molecular Neurobiology*, 55(7):5526–5536, Jul 2018. doi:10.1007/s12035-017-0758-1.

[25] Sarah L. Booth, M. Kyla Shea, Kathryn Barger, Sue E. Leurgans, Bryan D. James, Thomas M. Holland, Puja Agarwal, Xueyan Fu, Jifan Wang, Gregory Matuszek, and Julie A. Schneider. Association of vitamin k with cognitive decline and neuropathology in communitydwelling older persons. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 8, 01 2022. doi:10.1002/trc2.12255.

[26] Maria João Botelho, Jelena Milinovic, Narcisa M. Bandarra, and Carlos Vale. Alzheimer's disease and toxins produced by marine dinoflagellates: An issue to explore. *Marine Drugs*, 20, 04 2022. doi:10.3390/md20040253.

[27] Kiri L. Brickell, Ellen J. Steinbart, Malia Rumbaugh, Haydeh Payami, Gerard D. Schellenberg, Vivianna Van Deerlin, Wuxing Yuan, and Thomas D. Bird. Early-onset alzheimer disease in families with late-onset alzheimer disease. *Archives of Neurology*, 63, 09 2006. doi:10.1001/archneur.63.9.1307.

[28] Randy R. Brutkiewicz, Wei Cao, David Morgan, Roberta Souza Dos Reis, Vidyani Suryadevara, Auriel A. Willette, Sara A. Willette, Season K. Wyatt-Johnson, and Michael R. Duggan. What would it take to prove that a chronic infection is a causal agent in alzheimer's disease? *Trends in Neurosciences*, 48, 08 2025. doi:10.1016/j.tins.2025.05.009.

[29] Ryan Burslem, Kimberly Gottesman, Melanie Newkirk, and Jane Ziegler. Energy requirements for critically ill patients with covid19. *Nutrition in Clinical Practice*, 37, 06 2022. doi:10.1002/ncp.10852.

[30] Dana M. Cairns, Ruth F. Itzhaki, and David L. Kaplan. Potential involvement of varicella zoster virus in alzheimer's disease via reactivation of quiescent herpes simplex virus type 1. *Journal of Alzheimer's Disease*, 88, 08 2022. doi:10.3233/jad-220287.

[31] Alejandra Camacho-Soto, Irene Faust, Osvaldo J. Laurido-Soto, Jordan A. Killion, Natalie Senini, Brittany Krzyzanowski, and Brad A. Racette. Risk of developing alzheimer disease in relation to common infections. *Neurodegenerative Diseases*, 05 2025. doi:10.1159/000546589.

[32] Michael Camilleri. The aging gastrointestinal tract: digestive and motility problems, 04 2022. doi:10.1002/9781119600206.ch26.

[33] Ralph Carmel. Nutritional anemias and the elderly. *Seminars in Hematology*, 45(4):225–234, 2008. Anemia in the Elderly. URL: https://www.sciencedirect.com/science/article/pii/S003719630800125X, doi:10.1053/j.seminhematol.2008.07.009.

[34] Chris Carter. Alzheimerprimes disease: App, gamma secretase, apoe, clu, cr1, picalm, abca7, bin1, cd2ap, cd33, epha1, and ms4a2, and their relationships with herpes simplex, *c. pneumoniae*, other suspect pathogens, and the immune system. *International Journal of Alzheimer's Disease*, 2011, 01 2011. doi:10.4061/2011/501862.

[35] Federica Cavalcoli, Alessandra Zilli, Dario Conte, and Sara Massironi. Micronutrient deficiencies in patients with chronic atrophic autoimmune gastritis: A review., Jan 2017. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC5292330/pdf/WJG-23-563.pdf, doi:10.3748/wjg.v23.i4.563.

[36] Anu Chacko, Ali Delbaz, Heidi Walkden, Souptik Basu, Charles W. Armitage, Tanja Eindorf, Logan K. Trim, Edith Miller, Nicholas P. West, James A. St John, Kenneth W. Beagley, and Jenny K. Ekberg. Chlamydia pneumoniae can infect the central nervous system via the olfactory and trigeminal nerves and contributes to alzheimer's disease risk. *Scientific Reports*, 12, 02 2022. URL: https://www.nature.com/articles/s41598-022-06749-9, doi:10.1038/s41598-022-06749-9.

[37] Amitabh Chandra, Courtney Coile, and Corina Mommaerts. What can economics say about alzheimer's disease? *Journal of Economic Literature*, 61, 06 2023. doi:10.1257/jel.20211660.

[38] TsungHsien Chang, YingHsuan Tai, YingXiu Dai, YunTing Chang, TzengJi Chen, and MuHong Chen. Association between vitiligo and subsequent risk of dementia: A populationbased cohort study. *The Journal of Dermatology*, 48, 01 2021. doi:10.1111/1346-8138.15582.

[39] Julia KuUx017a[bad char vv=378]niar, Patrycja Kozubek, Magdalena Czaja, and Jerzy Leszek. Correlation between alzheimer's disease and gastrointestinal tract disorders. *Nutrients*, 16, 07 2024. doi:10.3390/nu16142366.

[40] Elissavet Chartampila, Karim S Elayouby, Paige Leary, John J LaFrancois, David Alcantara-Gonzalez, Swati Jain, Kasey Gerencer, Justin J Botterill, Stephen D Ginsberg, and Helen E Scharfman. Choline supplementation in early life improves and low levels of choline can impair outcomes in a mouse model of alzheimer's disease. *eLife*, 12:RP89889, jun 2024. doi:10.7554/eLife.89889.

[41] He Chen, Sixing Liu, Shuai Gao, Hangyu Shi, Yan Yan, Yixing Xu, Jiufei Fang, Weiming Wang, Huan Chen, and Zhishun Liu. Pharmacovigilance analysis of neurological adverse events associated with glp-1 receptor agonists based on the fda adverse event reporting system. *Scientific Reports*, 15, 05 2025. URL: https://www.nature.com/articles/s41598-025-01206-9, doi:10.1038/s41598-025-01206-9.

[42] Fu Cheng, Geng Peng, Yan Lu, Kang Wang, Qinuo Ju, Yongle Ju, and Manzhao Ouyang. Relationship between copper and immunity: The potential role of copper in tumor immunity. *Frontiers in Oncology*, 12, 11 2022. doi:10.3389/fonc.2022.1019153.

[43] Cezary Chojnacki, Anita GUx0105[bad char vv=261]siorowska, Tomasz PopUx0142[bad char vv=322]awski, Paulina Konrad, Marcin Chojnacki, Michal Fila, and Janusz Blasiak. Beneficial effect of increased tryptophan intake on its metabolism and mental state of the elderly. *Nutrients*, 15, 02 2023. doi:10.3390/nu15040847.

[44] Vlad Alexandru Ciurea, Razvan-Adrian Covache-Busuioc, Aurel George Mohan, Horia Petre Costin, and Victor Voicu. Alzheimer's disease: 120 years of research and progress., Feb 2023. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC10015576/, doi:10.25122/jml-2022-0111.

[45] Tara Conway, Karin Seidler, and Michelle Barrow. Unlocking choline's potential in alzheimer's disease: A narrative review exploring the neuroprotective and neurotrophic role of phosphatidylcholine and assessing its impact on memory and learning. *Clinical Nutrition ESPEN*, 64:177–195, 2024. URL: https://www.sciencedirect.com/science/article/pii/S2405457724013287, doi:10.1016/j.clnesp.2024.09.024.

[46] Janelle L. Cooper. P3400: Biotin deficiency and abnormal pantothenic acid levels in dementia. *Alzheimer's & Dementia*, 4, 07 2008. doi:10.1016/j.jalz.2008.05.1970.

[47] Marilyn Cornelis, Puja Agarwal, Thomas Holland, and Rob van Dam. Mind dietary pattern and its association with cognition and incident dementia in the uk biobank. *Nutrients*, 15, 01 2023. doi:10.3390/nu15010032.

[48] Tamer Coskun, Heidi K. Baumgartner, Shaoyou Chu, and Marshall H. Montrose. Coordinated regulation of gastric chloride secretion with both acid and alkali secretion. *American Journal of Physiology-Gastrointestinal and Liver Physiology*, 283, 11 2002. doi:10.1152/ajpgi.00184.2002.

[49] Paolina Crocco, Francesco De Rango, Serena Dato, Rossella La Grotta, Raffaele Maletta, Amalia Cecilia Bruni, Giuseppe Passarino, and Giuseppina Rose. The shortening of leukocyte telomere length contributes to alzheimer's disease: Further evidence from late-onset familial and sporadic cases. *Biology*, 12, 10 2023. doi:10.3390/biology12101286.

[50] B Cryer and M Feldman. Racial differences in gastric function among african americans and caucasian americans: secretion, serum gastrin, and histology. *Proceedings of the Association of American Physicians*, pages 481–9, Nov 1996. URL: https://pubmed.ncbi.nlm.nih.gov/8956372/.

[51] Jeffrey L. Cummings, Dana P. Goldman, Nicholas R. SimmonsStern, and Eric Ponton. The costs of developing treatments for alzheimer's disease: A retrospective exploration. *Alzheimer's & Dementia*, 18, 03 2022. doi:10.1002/alz.12450.

[52] Heng Dai and Ziyi Chen. Association between dietary vitamin k and telomere length: Based on nhanes 2001 to 2002. *Medicine*, 103, 2024. doi:10.1097/md.0000000000040157.

[53] Nikhil Dave, Jessica M. Judd, Annika Decker, Wendy Winslow, Patrick Sarette, Oscar Villarreal Espinosa, Savannah Tallino, Samantha K. Bartholomew, Alina Bilal, Jessica Sandler, Ian McDonough, Joanna K. Winstone, Erik A. Blackwood, Christopher Glembotski, Timothy Karr, and Ramon Velazquez. Dietary choline intake is necessary to prevent systemswide organ pathology and reduce alzheimer's disease hallmarks. *Aging Cell*, 22, 02 2023. doi:10.1111/acel.13775.

[54] Suzanne M. de la Monte and Ming Tong. Agent orange herbicidal toxin-initiation of alzheimer-type neurodegeneration. *Journal of Alzheimer's Disease*, 97, 02 2024. doi:10.3233/jad-230881.

[55] Akila De Silva and Stephen R Bloom. Gut hormones and appetite control: A focus on pyy and glp-1 as therapeutic targets in obesity. *Gut and liver*, pages 10–20, 01 2012. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC3286726/, doi:10.5009/gnl.2012.6.1.10.

[56] Rishika Dhapola, Prajjwal Sharma, Sneha Kumari, Jasvinder Singh Bhatti, and Dibbanti HariKrishnaReddy. Environmental toxins and alzheimer's disease: a comprehensive analysis of pathogenic mechanisms and therapeutic modulation. *Molecular Neurobiology*, 61(6):3657–3677, Jun 2024. doi:10.1007/s12035-023-03805-x.

[57] Ricardo Jorge Dinis-Oliveira. Ux201c[bad char vv=8220]not everything that can be counted countsux201d[bad char vv=8221] in ethanol toxicological results: an antemortem and postmortem technical interpretation focusing on driving under the influence. *Forensic Sciences Research*, 9, 07 2024. doi:10.1093/fsr/owae023.

[58] Hackwon Do, Nishanth Makthal, Arica R. VanderWal, Matthew Ojeda Saavedra, Randall J. Olsen, James M. Musser, and Muthiah Kumaraswami. Environmental ph and peptide signaling control virulence of streptococcus pyogenes via a quorum-sensing pathway. *Nature Communications*, 10, 06 2019. URL: https://www.nature.com/articles/s41467-019-10556-8, doi:10.1038/s41467-019-10556-8.

[59] Priyam Doshi, Lisbeth Escudero, Onose Okojie, Kimberly Miller, and Corey Sievers. S4128side effects of glp-1 agonists: A case of colonic pseudo-obstruction. *American Journal of Gastroenterology*, 120, 2025. doi:10.14309/01.ajg.0001143972.38422.8d.

[60] Gerald W. Dryden and Stephen A. McClave. *Gastrointestinal Senescence and Digestive Diseases of the Elderly*, pages 569–581. Humana Press, Totowa, NJ, 2004. doi:10.1007/978-1-59259-391-0_25.

[61] Alex Eble and Feng Hu. Signals, information, and the value of college names. *Review of Economics and Statistics*, 107. doi:10.1162/rest_a_01277.

[62] Paul Edison, Grazia Daniela Femminella, Craig Ritchie, Joseph Nowell, Clive Holmes, Zuzana Walker, Basil Ridha, Sanara Raza, Nicholas R. Livingston, Eleni Frangou, Sharon Love, Gareth Williams, Robert Lawrence, Brady Mcfarlane, Hilary Archer, Elizabeth Coulthard, Benjamin R. Underwood, Paul Koranteng, Salman Karim, Carol Bannister, Robert Perneczky, Aparna Prasanna, Kehinde Junaid, Bernadette McGuinness, Ramin Nilforooshan, Ajay Macharouthu, Andrew Donaldson, Simon Thacker, Gregor Russell, Naghma Malik, Vandana Mate, Lucy Knight, Sajeev Kshemendran, Christian Holscher, Anita Mansouri, Mae Chester-Jones, Jane Holmes, Trisha Tan, Steve Williams, Azhaar Ashraf, David J. Brooks, John Harrison, Rainer Hinz, George Tadros, Anthony Peter Passmore, and Clive Ballard. Liraglutide in mild to moderate alzheimer's disease: a phase 2b clinical trial. *Nature Medicine*, 12 2025. URL: https://www.nature.com/articles/s41591-025-04106-7, doi:10.1038/s41591-025-04106-7.

[63] Ebru Emekli-Alturfan and A. Ata Alturfan. The emerging relationship between vitamin k and neurodegenerative diseases: a review of current evidence. *Molecular Biology Reports*, 50(1):815–828, Jan 2023. doi:10.1007/s11033-022-07925-w.

[64] Ellen Fallows. Malnutrition with use of glp-1 agonists is an underestimated real world harm. *BMJ*, 390, 07 2025. doi:10.1136/bmj.r1512.

[65] Jeffrey Fessel. Supplemental thiamine as a practical, potential way to prevent alzheimer's disease from commencing. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 7, 01 2021. doi:10.1002/trc2.12199.

[66] Orestes V. Forlenza, Vanessa J. de Paula, Rodrigo Machado-Vieira, Breno S. Diniz, and Wagner F. Gattaz. Does lithium prevent alzheimer's disease? *Drugs & Aging*, 29(5):335–342, May 2012. doi:10.2165/11599180-000000000-00000.

[67] Orestes Vicente Forlenza and Breno José Alencar Pires Barbosa. What are the reasons for the repeated failures of clinical trials with anti-amyloid drugs for ad treatment? *Dementia & Neuropsychologia*, 19, 2025. doi:10.1590/1980-5764-dn-2025-e001.

[68] Sanne Franzen, Paola Gilsanz, Wei Li, Alison J. McManus, Debora Melo van Lent, Sadaf Arefi Milani, C. Elizabeth Shaaban, Shana D. Stites, Erin Sundermann, Vidyani Suryadevara, JeanFrancoise Trani, Arlener D. Turner, Jet M. J. Vonk, Yakeel T. Quiroz, Ganesh M. Babulal, Michelle M. Mielke, Neelum T. Aggarwal, Clara VilaCastelar, Puja Agarwal, Eider M. ArenazaUrquijo, Benjamin Brett, Anna BrugulatSerrat, Lyndsey E. DuBose, Willem S. Eikelboom, Jason Flatt, and Nancy S. Foldi. Consideration of sex and gender in alzheimer's disease and related disorders from a global perspective. *Alzheimer's & Dementia*, 18, 12 2022. doi:10.1002/alz.12662.

[69] Hershey H. Friedman. The education irony: when college degrees lead to unemployment, mindless thinking, debt, and despair. *Academia Mental Health and Well-Being*, 2, 04 2025. doi:10.20935/mhealthwellb7661.

[70] Ai Ling Fu, Qian Li, Zhao Hui Dong, Shi Jie Huang, Yu Xia Wang, and Man Ji Sun. Alternative therapy of alzheimer's disease via supplementation with choline acetyltransferase. *Neuroscience Letters*, 368(3):258–262, 2004. URL: https://www.sciencedirect.com/science/article/pii/S0304394004007153, doi:10.1016/j.neulet.2004.05.116.

[71] Kanako Fujii, Toshihide Takeuchi, Yuzo Fujino, Noriko Tanaka, Nao Fujino, Akiko Takeda, Eiko N. Minakawa, and Yoshitaka Nagai. Oral administration of arginine suppresses aupbeta pathology in animal models of alzheimer's disease. *Neurochemistry International*, 191, 12 2025. doi:10.1016/j.neuint.2025.106082.

[72] Cory C. Funk, Tom Paterson, Alex Bangs, David M. Cannon, George Savage, Eric Ringger, and Lee Hood. Mining the gaps: Deciphering alzheimer's biology through ai-driven reconciliation. *The Journal of Prevention of Alzheimer's Disease*, 13(1):100402, 2026. URL: https://www.sciencedirect.com/science/article/pii/S2274580725003449, doi:

10.1016/j.tjpad.2025.100402.

[73] Serena Galiè, Silvia Canudas, Jananee Muralidharan, Jesús García-Gavilán, Mònica Bulló, and Jordi Salas-Salvadó. Impact of nutrition on telomere health: Systematic review of observational cohort studies and randomized clinical trials. *Advances in Nutrition*, 11(3):576–601, 2020. URL: https://www.sciencedirect.com/science/article/pii/S2161831322002861, doi:10.1093/advances/nmz107.

[74] Tal Ganz, Nina Fainstein, and Tamir Ben-Hur. When the infectious environment meets the ad brain. *Molecular Neurodegeneration*, 17(1):53, Aug 2022. doi:10.1186/s13024-022-00559-3.

[75] Song Gao, Edward C. Bell, Yun Zhang, and Dong Liang. Racial disparity in drug disposition in the digestive tract. *International Journal of Molecular Sciences*, 22, 02 2021. doi:10.3390/ijms22031038.

[76] Giacomo Garibotto, Paolo Tessari, Daniela Verzola, and Laura Dertenois. The metabolic conversion of phenylalanine into tyrosine in the human kidney: Does it have nutritional implications in renal patients? *Journal of Renal Nutrition*, 12, 01 2002. doi:10.1053/jren.2002.29600.

[77] Stephen J. Genuis and Kasie L. Kelln. Toxicant exposure and bioaccumulation: A common and potentially reversible cause of cognitive dysfunction and dementia. *Behavioural Neurology*, 2015, 2015. doi:10.1155/2015/620143.

[78] Markus Goldschmiedt, Cora C. Barnett, Barry E. Schwarz, William E. Karnes, Jan S. Redfern, and Mark Feldman. Effect of age on gastric acid secretion and serum gastrin concentrations in healthy men and women. *Gastroenterology*, 101(4):977–990, 1991. URL: https://www.sciencedirect.com/science/article/pii/001650859190724Y, doi:10.1016/0016-5085(91)90724-Y.

[79] Lorenzo Grimaldi, Rosaria A. Cavallaro, Domenico De Angelis, Andrea Fuso, and Giulia Sancesario. Vitamin k properties in stroke and alzheimer's disease: A janus bifrons in protection and prevention. *Molecules*, 30, 03 2025. doi:10.3390/molecules30051027.

[80] Corlia Grobler, Marvi van Tongeren, Jan Gettemans, Douglas B. Kell, and Etheresia Pretorius. Alzheimer's disease: A systems view provides a unifying explanation of its development. *Journal of Alzheimer's Disease*, 91, 01 2023. doi:10.3233/jad-220720.

[81] Jiang Haowei, Newman Morgan, and Michael Lardelli. The zebrafish orthologue of familial alzheimer's disease gene presenilin 2 is required for normal adult melanotic skin pigmentation. *PLOS ONE*, 13(10):1–20, 10 2018. doi:10.1371/journal.pone.0206155.

[82] A J Harwood. Lithium and bipolar mood disorder: the inositol-depletion hypothesis revisited. *Molecular Psychiatry*, 10, 01 2005. URL: https://www.nature.com/articles/4001618, doi:10.1038/sj.mp.4001618.

[83] Zaw Myo Hein, Barani Karikalan, Prarthana Kalerammana Gopalakrishna, Krina Dhevi, Aisyah Alkatiri, Farida Hussan, Mohamad Aris Mohd Moklas, Saravanan Jagadeesan, Muhammad Danial Che Ramli, Che Nasril Che Mohd Nassir, and Thirupathirao Vishnumukkala. Toward a unified framework in molecular neurobiology of alzheimer's disease: Revisiting the pathophysiological hypotheses. *Molecular Neurobiology*, 63(1):282, Dec 2025. doi:10.1007/s12035-025-05602-0.

[84] Virgilio Hernandez-Ruiz, Luc Letenneur, Tamas Fülöp, Catherine Helmer, Claire Roubaud-Baudron, José-Alberto Avila-Funes, and Hélène Amieva. Infectious diseases and cognition: do we have to worry? *Neurological Sciences*, 43(11):6215–6224, Nov 2022. doi:10.1007/s10072-022-06280-9.

[85] Fenna Hinssen, Marco Mensink, Thom Huppertz, and Nikkie van der Wielen. Impact of aging on the digestive system related to protein digestion in vivo. *Critical Reviews in Food Science and Nutrition*, 65, 10 2025. doi:10.1080/10408398.2024.2433598.

[86] Peter R. Holt. Intestinal malabsorption in the elderly. *Digestive Diseases*, 25, 04 2007. doi:10.1159/000099479.

[87] Muhammad Kamal Hossain and Hyung-Ryong Kim. Taurine as an early-phase disease-modifying candidate for alzheimer's disease. *International journal of molecular sciences*, 02 2026. URL: https://pubmed.ncbi.nlm.nih.gov/41752008/, doi:10.3390/ijms27041871.

[88] Yujun Hou, Yong Wei, Sofie Lautrup, Beimeng Yang, Yue Wang, Stephanie Cordonnier, Mark P Mattson, Deborah L Croteau, and Vilhelm A Bohr. Nad(+) supplementation reduces neuroinflammation and cell senescence in a transgenic mouse model of alzheimer's disease via cgas-sting. *Proceedings of the National Academy of Sciences of the United States of America*, Sep 2021. URL: https://pubmed.ncbi.nlm.nih.gov/34497121/, doi:10.1073/pnas.2011226118.

[89] Ya Hui Hung, Elysia L Robb, Irene Volitakis, Michael Ho, Genevieve Evin, Qiao-Xin Li, Janetta G Culvenor, Colin L Masters, Robert A Cherny, and Ashley I Bush. Paradoxical condensation of copper with elevated beta-amyloid in lipid rafts under cellular copper deficiency conditions: implications for alzheimer disease. *The Journal of biological chemistry*, pages 21899–21907, 06 2009. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC2755914/, doi:10.1074/jbc.M109.019521.

[90] Aryeh Hurwitz. Gastric acidity in older adults. *JAMA: The Journal of the American Medical Association*, 278, 08 1997. doi:10.1001/jama.1997.03550080069041.

[91] Vojtechova Iveta, Machacek Tomas, Kristofikova Zdenka, Stuchlik Ales, and Tomas Petrasek. Infectious origin of alzheimer's disease: Amyloid beta as a component of brain antimicrobial immunity. *PLOS Pathogens*, 18(11):1–25, 11 2022. doi:10.1371/journal.ppat.1010929.

[92] Nina G. Jablonski and George Chaplin. Human skin pigmentation, migration and disease susceptibility. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 367, 03 2012. doi:10.1098/rstb.2011.0308.

[93] Ruchi Jakhmola-Mani, Anam Islam, and Deepshikha Pande Katare. Liver-brain axis in sporadic alzheimer's disease: Role of ten signature genes in a mouse model. *CNS & Neurological Disorders - Drug Targets*, 20, 11 2021. doi:10.2174/1871527319666201209111006.

[94] Angelo M. Jamerlan, Seong Soo A. An, and John P. Hulme. Microbial diversity and fitness in the gut-brain axis: influences on developmental risk for alzheimer's disease. *Gut Microbes*, 17, 12 2025. doi:10.1080/19490976.2025.2486518.

[95] Narumol Jariyasopit, Tiwat Phochmak, Siriphan Manocheewa, Kwanjeera Wanichthanarak, Suphitcha Limjiasahapong,

Nichapa Kleebkomut, Yongyut Sirivatanauksorn, Vorapan Sirivatanauksorn, Arintaya Phrommintikul, Nipon Chattipakorn, Siriporn Chattipakorn, and Sakda Khoomrung. Higher plasma kynurenine to tryptophan correlates with an increased incidence of mild cognitive impairment in treated metabolic syndrome patients. *ACS Omega*, 10(51):63226–63237, 2025. arXiv:https://doi.org/10.1021/acsomega.5c09713, doi:10.1021/acsomega.5c09713.

[96] Varuna H. Jasodanand, Matteo Bellitti, and Vijaya B. Kolachalama. An aifirst framework for multimodal data in alzheimer's disease and related dementias. *Alzheimer's & Dementia*, 21, 09 2025. doi:10.1002/alz.70719.

[97] Xinchen Ji, Jian Wang, Tianye Lan, Dexi Zhao, and Peng Xu. Gut microbial metabolites and the brain-gut axis in alzheimer's disease: A review. *Biomolecules & biomedicine*, pages 240–250, 08 2025. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC12505531/, doi:10.17305/bb.2025.12921.

[98] Alan Wayne Jones. Comment on 'estimates of non-alcoholic food-derived ethanol and methanol in human'. *Journal of Analytical Toxicology*, 46(1):e48–e51, 10 2021. arXiv:https://academic.oup.com/jat/article-pdf/46/1/e48/42508144/bkab112.pdf, doi:10.1093/jat/bkab112.

[99] Jessica M Judd, Faizan Mistry, Wendy Winslow, Savannah Tallino, Julie Turk, and Ramon Velazquez. The 3xtg-ad mouse model of alzheimer's disease exhibits lifelong reductions in circulating choline despite adequate dietary intake, with sex-specific neuropathological and behavioral phenotypes. *Aging cell*, page e70330, Jan 2026. URL: https://pubmed.ncbi.nlm.nih.gov/41457719/, doi:10.1111/acel.70330.

[100] Joon Hyung Jung, Min Soo Byun, Dahyun Yi, Hyejin Ahn, Jun Ho Lee, Jang-Seok Lee, Hyun-Seob Lee, Jun-Young Lee, Yu Kyeong Kim, Yun-Sang Lee, Koung Mi Kang, Chul-Ho Sohn, and Dong Young Lee. Telomere length, in vivo alzheimer's disease pathologies and cognitive decline in older adults. *Journal of Neurology, Neurosurgery & Psychiatry*, 96(6):558–565, 2025. URL: https://jnnp.bmj.com/content/96/6/558, arXiv:https://jnnp.bmj.com/content/96/6/558.full.pdf, doi:10.1136/jnnp-2024-334314.

[101] Gwendolyn E. Kaeser and Jerold Chun. Mosaic somatic gene recombination as a potentially unifying hypothesis for alzheimer's disease. *Frontiers in Genetics*, 11, 05 2020. doi:10.3389/fgene.2020.00390.

[102] Joanna Kaluzna-Czaplinska, Paulina Gatarek, , Salvatore Chirumbolo, Max Stanley Chartrand, and Geir Bjorklund. How important is tryptophan in human health? *Critical Reviews in Food Science and Nutrition*, 59, 2017. doi:10.1080/10408398.2017.1357534.

[103] Julia Kamierczak-Baraska and Bolesaw T Karwowski. The protective role of vitamin k in aging and age-related diseases. *Nutrients*, 12 2024. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC11676630/, doi:10.3390/nu16244341.

[104] Sohye Kang, Johanna Mullen, Les P. Miranda, and Rohini Deshpande. Utilization of tyrosine and histidinecontaining dipeptides to enhance productivity and culture viability. *Biotechnology and Bioengineering*, 109, 09 2012. doi:10.1002/bit.24507.

[105] Chrysoula-Evangelia Karachaliou and Evangelia Livaniou. Biotin homeostasis and human disorders: Recent findings and perspectives. *International Journal of Molecular Sciences*, 25, 06 2024. doi:10.3390/ijms25126578.

[106] Nicholas Karagas, Jessica E. Young, Elizabeth E. Blue, and Suman Jayadev. The spectrum of genetic risk in alzheimer disease. *Neurology Genetics*, 11, 02 2025. doi:10.1212/nxg.0000000000200224.

[107] Chenmala Karthika, Anoop Pattanoor Appu, Rokeya Akter, Md. Habibur Rahman, Priti Tagde, Ghulam Md. Ashraf, Mohamed M. Abdel-Daim, Syed ul Hassan, Areha Abid, and Simona Bungau. Potential innovation against alzheimer's disorder: a tricomponent combination of natural antioxidants (vitamin e, quercetin, and basil oil) and the development of its intranasal delivery. *Environmental Science and Pollution Research*, 29(8):10950–10965, Feb 2022. doi:10.1007/s11356-021-17830-7.

[108] Takele Adugna Kassegn, Zhancheng Tian, Junzheng Du, Qingli Niu, Jifei Yang, Zhang Hongge, Yuqing Song, Fengxue Chai, Zhonghui Zhang, Guiquan Guan, and Hong Yin. Nad(+) homeostasis attenuates japanese encephalitis virus infection progression. *FASEB journal : official publication of the Federation of American Societies for Experimental Biology*, page e71526, Feb 2026. URL: https://pubmed.ncbi.nlm.nih.gov/41693302/, doi:10.1096/fj.202503827R.

[109] Timothy L. Kauffman, Ronald W. Scott, John O. Barr, and Michael L. Moran. *A Comprehensive Guide to Geriatric Rehabilitation E-Book*. Elsevier Health Sciences, 09 2014. URL: https://www.google.com/books/edition/A_Comprehensive_Guide_to_Geriatric_Rehab/UDwPBAAAQBAJ?hl=en&gbpv=1&dq=digestive+malabsorption+elderly+&pg=PA45&printsec=frontcover.

[110] Elisa Mitiko Kawamoto, Carmen Vivar, and Simonetta Camandola. Physiology and pathology of calcium signaling in the brain. *Frontiers in Pharmacology*, 3, 2012. doi:10.3389/fphar.2012.00061.

[111] Alan D Kaye, Austin S Thomassen, Sydney A Mashaw, Ellie M MacDonald, Aubert Waguespack, Lily Hickey, Anushka Singh, Deniz Gungor, Anusha Kallurkar, Adam M Kaye, Sahar Shekoohi, and Giustino Varrassi. Vitamin e (upalpha-tocopherol): Emerging clinical role and adverse risks of supplementation in adults. *Cureus*, 02 2025. doi:10.7759/cureus.78679.

[112] M. Kekki, I. M. Samloff, T. Ihamäki, K. Varis, and M. Siurala. Age- and sex-related behaviour of gastric acid secretion at the population level. *Scandinavian Journal of Gastroenterology*, 17. doi:10.3109/00365528209181087.

[113] Katherine M. Kennedy, Marcus C. de Goffau, Maria Elisa Perez-Muñoz, Marie-Claire Arrieta, Fredrik Bäckhed, Peer Bork, Thorsten Braun, Frederic D. Bushman, Joel Dore, Willem M. de Vos, Ashlee M. Earl, Jonathan A. Eisen, Michal A. Elovitz, Stephanie C. Ganal-Vonarburg, Michael G. Gänzle, Wendy S. Garrett, Lindsay J. Hall, Mathias W. Hornef, Curtis Huttenhower, Liza Konnikova, Sarah Lebeer, Andrew J. Macpherson, Ruth C. Massey, Alice Carolyn McHardy, Omry Koren, Trevor D. Lawley, Ruth E. Ley, Liam O'Mahony, Paul W. O'Toole, Eric G. Pamer, Julian Parkhill, Jeroen Raes, Thomas Rattei, Anne Salonen, Eran Segal, Nicola Segata, Fergus Shanahan, Deborah M. Sloboda, Gordon S. Smith, Harry Sokol, Tim D. Spector, Michael G. Surette, Gerald W. Tannock, Alan W. Walker, Moran Yassour, and Jens Walter. Questioning the fetal microbiome illustrates pitfalls of low-biomass microbial studies. *Nature*, 613, 01 2023.

URL: https://www.nature.com/articles/s41586-022-05546-8, doi:10.1038/s41586-022-05546-8.

[114] Masahiro Kita, Kuniaki Obara, Sumio Kondo, Satoshi Umeda, and Yasuhisa Ano. Effect of supplementation of a whey peptide rich in tryptophan-tyrosine-related peptides on cognitive performance in healthy adults: A randomized, double-blind, placebo-controlled study. *Nutrients*, 10, 07 2018. doi:10.3390/nu10070899.

[115] Jule Klotter. Niacinamide (nicotinamide) and alzheimer's disease. *Townsend Letter*, 12 2010. URL: https://go.gale.com/ps/i.do?id=GALE%7CA243290423&sid=googleScholar&v=2.1&it=r&linkaccess=abs&issn=19405464&p=HRCA&sw=w&userGroupName=anon%7E7071297&aty=open-web-entry.

[116] Martin Kohlmeier, Andreas Salomon, Jörg Saupe, and Martin J. Shearer. Transport of vitamin k to bone in humans. *The Journal of Nutrition*, 126:1192S–1196S, 1996. URL: https://www.sciencedirect.com/science/article/pii/S0022316622017527, doi:10.1093/jn/126.suppl_4.1192S.

[117] Piotr Konopelski and Izabella Mogilnicka. Biological effects of indole-3-propionic acid, a gut microbiota-derived metabolite, and its precursor tryptophan in mammals' health and disease. *International Journal of Molecular Sciences*, 23, 02 2022. doi:10.3390/ijms23031222.

[118] Xi-Yuen Kuan, Nurul Ahmad Fauzi, Khuen Yen Ng, and Athirah Bakhtiar. Exploring the causal relationship between telomere biology and alzheimer's disease. *Molecular Neurobiology*, 60(8):4169–4183, Aug 2023. doi:10.1007/s12035-023-03337-4.

[119] Satoshi Kurisu and Hitoshi Fujiwara. Overestimation of aortic stenosis in thiamine deficiency: Two cases with distinct underlying conditions. *Cureus*, page e99487, 12 2025. URL: https://pubmed.ncbi.nlm.nih.gov/41552235/, doi:10.7759/cureus.99487.

[120] Maiko Kusano, Chikara Kohda, Masaya Fujishiro, and Taka-aki Matsuyama. Ethanol production in the gut: an autopsy case. *Journal of Analytical Toxicology*, 49(6):422–426, 05 2025. arXiv:https://academic.oup.com/jat/article-pdf/49/6/422/63056819/bkaf039.pdf, doi:10.1093/jat/bkaf039.

[121] Carolina Lajterer, Carmit Shani Levi, and Uri Lesmes. An in vitro digestion model accounting for sex differences in gastro-intestinal functions and its application to study differential protein digestibility. *Food Hydrocolloids*, 132:107850, 2022. URL: https://www.sciencedirect.com/science/article/pii/S0268005X22003708, doi:10.1016/j.foodhyd.2022.107850.

[122] Sofie Lautrup, David A Sinclair, Mark P Mattson, and Evandro F Fang. Nad(+) in brain aging and neurodegenerative disorders. *Cell metabolism*, pages 630–655, Oct 2019. URL: https://pubmed.ncbi.nlm.nih.gov/31577933/, doi:10.1016/j.cmet.2019.09.001.

[123] Johanna M. S. Lemons, Jenni Firrman, Karley K. Mahalak, LinShu Liu, Adrienne B. Narrowe, Stephanie Higgins, Ahmed M. Moustafa, Aurélien Baudot, Stef Deyaert, and Pieter Van den Abbeele. The effect of sodium benzoate on the gut microbiome across age groups. *Foods*, 14, 09 2025. doi:10.3390/foods14172949.

[124] Hongwei Li, Changhua Shi, Keya Li, Xinjing Fu, Ying Lyu, Yanfeng Xu, Yunlin Han, Wei Liang, Qin Chuan, and Ling Zhang. Chronic methanol exposure induces cognitive impairment and alzheimer's-like pathology in rhesus monkeys. *Animal Models and Experimental Medicine*, 02 2026. URL: https://onlinelibrary.wiley.com/doi/full/10.1002/ame2.70142, doi:10.1002/ame2.70142.

[125] Jinhe Li, Huilin Mou, and Ranran Yao. Gut-brain axis and a systematic approach to alzheimer's disease therapies. *Brain Science Advances*, 11, 01 2025. doi:10.26599/bsa.2024.9050031.

[126] S W Li, M Arita, A Fertala, Y Bao, G C Kopen, T K Lngsj, M M Hyttinen, H J Helminen, and D J Prockop. Transgenic mice with inactive alleles for procollagen n-proteinase (adamts-2) develop fragile skin and male sterility. *The Biochemical journal*, pages 271–8, Apr 2001. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC1221736/, doi:10.1042/0264-6021:3550271.

[127] Zuguang Li, Juan Zhang, Zhiqiang Liu, Lu Yu, Chunqing Yang, Luoman Zhang, Zhigao Xiang, Feng Hu, Nadezda Brazh, Kai Shu, Ling-Qiang Zhu, and Dan Liu. Brain somatic mutations in alzheimer's disease: linking genetic mosaicism to neurodegeneration. *Molecular Neurodegeneration*, 20(1):104, Oct 2025. doi:10.1186/s13024-025-00895-0.

[128] Chun-Wei Liang, Hsiao-Yi Cheng, and Mei-Chih Meg Tseng. Effects of sodium benzoate on cognitive function in neuropsychiatric disorders: a systematic review and meta-analysis. *Frontiers in Psychiatry*, 15, 09 2024. doi:10.3389/fpsyt.2024.1370431.

[129] Ru-jia Liao, Lei Jiang, Rong-rong Wang, Hua-wei Zhao, Ying Chen, Ya Li, Lu Wang, Li-Yong Jie, Yu-dong Zhou, Xiang-nan Zhang, Zhong Chen, and Wei-wei Hu. Histidine provides long-term neuroprotection after cerebral ischemia through promoting astrocyte migration. *Scientific Reports*, 5, 10 2015. URL: https://www.nature.com/articles/srep15356, doi:10.1038/srep15356.

[130] Unhee Lim, Songren Wang, SongYi Park, David Bogumil, Anna H. Wu, Iona Cheng, Christopher A. Haiman, Loïc Le Marchand, Lynne R. Wilkens, Lon White, and V. Wendy Setiawan. Risk of alzheimer's disease and related dementia by sex and race/ethnicity: The multiethnic cohort study. *Alzheimer's & Dementia*, 18, 09 2022. doi:10.1002/alz.12528.

[131] Bowen Lin, Yu Hasegawa, Koki Takane, Nobutaka Koibuchi, Cheng Cao, and Shokei KimMitsuyama. High‐fat‐diet intake enhances cerebral amyloid angiopathy and cognitive impairment in a mouse model of alzheimer's disease, independently of metabolic disorders. *Journal of the American Heart Association*, 5(6):e003154, 2016. URL: https://www.ahajournals.org/doi/abs/10.1161/JAHA.115.003154, arXiv:https://www.ahajournals.org/doi/pdf/10.1161/JAHA.115.003154, doi:10.1161/JAHA.115.003154.

[132] Chieh-Hsin Lin, Ping-Kun Chen, Shi-Heng Wang, and Hsien-Yuan Lane. Effect of sodium benzoate on cognitive function among patients with behavioral and psychological symptoms of dementia. *JAMA Network Open*, 4, 04 2021. doi:10.1001/jamanetworkopen.2021.6156.

[133] ChiehHsin Lin, ShiHeng Wang, and HsienYuan Lane. Sodium benzoate, a damino acid oxidase inhibitor, improved

shortterm memory in patients with mild cognitive impairment in a randomized, doubleblind, placebocontrolled clinical trial. *Psychiatry and Clinical Neurosciences*, 79, 08 2025. doi:10.1111/pcn.13841.

[134] Jiankang Liu and Bruce N. Ames. Reducing mitochondrial decay with mitochondrial nutrients to delay and treat cognitive dysfunction, alzheimer's disease, and parkinson's disease. *Nutritional Neuroscience*, 8. doi:10.1080/10284150500047161.

[135] Yujie Lyu, Zhipeng Meng, Yunyun Hu, Bing Jiang, Jiao Yang, Yiqin Chen, Jun Zhou, Mingcheng Li, and Huping Wang. Mechanisms of mitophagy and oxidative stress in cerebral ischemia-reperfusion, vascular dementia, and alzheimer's disease. *Frontiers in Molecular Neuroscience*, 17, 08 2024. doi:10.3389/fnmol.2024.1394932.

[136] Xin Ma, Xin-Meng Wang, Guo-Zhang Tang, Yi Wang, XueUx2011[bad vv=8209]Chun Liu, Shuai-Deng Wang, Peng Peng, Xiu-Hong Qi, Xin-Ya Qin, YueUx2011[bad vv=8209]Ju Wang, Chen-Wei Wang, and Jiang-Ning Zhou. Alterations of amino acids in older adults with alzheimer's disease and vascular dementia. *Amino Acids*, 57(1):10, Jan 2025. doi:10.1007/s00726-024-03442-1.

[137] Stefania Maggi, Tamàs Fulöp, Elda De Vita, Federica Limongi, Damiano Pizzol, Francesco Di Gennaro, and Nicola Veronese. Association between vaccinations and risk of dementia: a systematic review and meta-analysis. *Age and Ageing*, 54(11):afaf331, 11 2025. arXiv:https://academic.oup.com/ageing/article-pdf/54/11/afaf331/65450501/afaf331.pdf, doi:10.1093/ageing/afaf331.

[138] Timoth J. Maher, Paul J. Kiritsy, Fernando A. Moya-Huff, Franco Casacci, Franco De Marchi, and Richard J. Wurtman. Use of parenteral dipeptides to increase serum tyrosine levels and to enhance catecholamine-mediated neurotransmission. *Journal of Pharmaceutical Sciences*, 79(8):685–687, 1990. URL: https://www.sciencedirect.com/science/article/pii/S0022354915483078, doi:10.1002/jps.2600790807.

[139] Gunjan D. Manocha, Angela M. Floden, Nicole M. Miller, Abbie J. Smith, Kumi Nagamoto-Combs, Takashi Saito, Takaomi C. Saido, and Colin K. Combs. Temporal progression of alzheimer's disease in brains and intestines of transgenic mice. *Neurobiology of Aging*, 81:166–176, 2019. URL: https://www.sciencedirect.com/science/article/pii/S0197458019301812, doi:10.1016/j.neurobiolaging.2019.05.025.

[140] Mohamed Ezzat M. Mansour, Ahmed Hamdy G. Ali, Mohamed Hazem M. Ibrahim, Ahella Ismail A. Mousa, and Ahmed Said Negida. Safety and efficacy of sodium benzoate for patients with mild alzheimer's disease: a systematic review and meta-analysis. *Nutritional Neuroscience*, 28, 06 2025. doi:10.1080/1028415x.2024.2415867.

[141] Mike J Marchywka. Vitamin d: Towards a conflict resolving hypothesis. techreport MJM-2022-010, not institutionalized, independent, 306 Charles Cox , Canton GA 30115, 7 2022. URL: https://www.researchgate.net/publication/362303875_Vitamin_D_Towards_A_Conflict_Resolving_Hypothesis, doi:10.5281/zenodo.6922402.

[142] Mike J Marchywka. Brain microbiome : Make your garden grow? feed your head. techreport MJM-2023-008, not institutionalized, independent, 44 Crosscreek Trail, Jasper GA 30143, 10 2023. URL: https://www.researchgate.net/publication/374946157_potuse-2023-10-24-v3, doi:10.5281/zenodo.10038912.

[143] Mike J Marchywka. Copper supplementation in dogs: Listen to her heart everybody's got a hungry heart. techreport MJM-2024-010, not institutionalized, independent, 157 Zachary Drive Talking Rock GA 30175 GA, 7 2025. URL: https://www.researchgate.net/publication/393240388_Copper_Supplementation_in_Dogs, doi:10.5281/zenodo.15785347.

[144] M.J. Marchywka. On the age distribution of sars-cov-2 patients. Technical Report MJM-2020-002-0.12, not institutionalized , independent, 306 Charles Cox , Canton GA 30115, 3 2021. Version 0.10 , may change signficantly if less than 1.00. URL: https://www.researchgate.net/publication/350021196_On_the_age_distribution_of_covid-19_patients.

[145] Fawziah Marra, Kyle Gomes, Emily Liu, Nirma Khatri Vadlamudi, Kathryn Richardson, and Jacquelyn J. Cragg. Effects of herpes zoster infection, antivirals and vaccination on risk of developing dementia: A systematic review and meta-analysis. *Human Vaccines & Immunotherapeutics*, 21, 12 2025. doi:10.1080/21645515.2025.2546741.

[146] Paula Martinez-Feduchi, Peng Jin, and Bing Yao. Epigenetic modifications of dna and rna in alzheimer's disease. *Frontiers in Molecular Neuroscience*, 17, 04 2024. doi:10.3389/fnmol.2024.1398026.

[147] Mohamad Norouzi Masir and Milad Shirvaliloo. Symptomatology and microbiology of the gastrointestinal tract in <scp>postcovid</scp> conditions. *JGH Open*, 6, 10 2022. doi:10.1002/jgh3.12811.

[148] Shinji Matsunaga, Taro Kishi, Peter Annas, Hans Basun, Harald Hampel, and Nakao Iwata. Lithium as a treatment for alzheimer's disease: A systematic review and meta-analysis. *Journal of Alzheimer's Disease*, 48, 09 2015. doi:10.3233/jad-150437.

[149] Louis D Matzel, Yu Han, Mauricio Lavie, and Chetan C Gandhi. Calcium 'leak' through somatic l-type channels has multiple deleterious effects on regulated transmitter release from an invertebrate hair cell. *Brain research*, pages 9–20, Mar 2003. URL: https://pubmed.ncbi.nlm.nih.gov/12591115/, doi:10.1016/s0006-8993(02)03883-0.

[150] I. M. Mazurkiewicz-Kwilecki and S. Nsonwah. Changes in the regional brain histamine and histidine levels in postmortem brains of alzheimer patients. *Canadian Journal of Physiology and Pharmacology*, 67, 01 1989. doi:10.1139/y89-013.

[151] Babacar Mbaye, Reham Magdy Wasfy, Maryam Tidjani Alou, Patrick Borentain, Rene Gerolami, Jean-Charles Dufour, and Matthieu Million. A catalog of ethanol-producing microbes in humans. *Future Microbiology*, 19, 05 2024. doi:10.2217/fmb-2023-0250.

[152] Elizabeth A McAninch, Kumar B Rajan, Denis A Evans, Sungro Jo, Layal Chaker, Robin P Peeters, David A Bennett, Deborah C Mash, and Antonio C Bianco. A common dio2 polymorphism and alzheimer disease dementia in african and european americans. *The Journal of Clinical Endocrinology & Metabolism*, 103(5):1818–1826, 02 2018. arXiv:https://academic.oup.com/jcem/article-pdf/103/5/1818/25088334/jc.2017-01196.pdf, doi:10.1210/jc.2017-01196.

[153] Seyyed Sam Mehdi Hosseininasab, Rasoul Ebrahimi, Shirin Yaghoobpoor, Kiarash Kazemi, Yaser Khakpour, Ramtin Hajibeygi, Ashraf Mohamadkhani, Mobina Fathi, Kimia Vakili, Arian Tavasol, Zohreh Tutunchian, Tara Fazel, Mohammad Fathi, and Mohammadreza Hajiesmaeili. Alzheimer's disease and infectious agents: a comprehensive review of pathogenic mechanisms and microrna roles. *Frontiers in Neuroscience*, 18, 01 2025. doi:10.3389/fnins.2024.1513095.

[154] Raaj S Mehta, Bharati Kochar, Zhen Zhou, Jonathan C Broder, Paget Chung, Keming Yang, Jessica Lockery, Michelle Fravel, Joanne Ryan, Suzanne Mahady, Suzanne G Orchard, John J McNeil, Anne Murray, Robyn L Woods, Michael E Ernst, and Andrew T Chan. Association of proton pump inhibitor use with incident dementia and cognitive decline in older adults: A prospective cohort study. *Gastroenterology*, pages 564–572.e1, 06 2023. URL: https://pubmed.ncbi.nlm.nih.gov/37315867/, doi:10.1053/j.gastro.2023.05.052.

[155] Abraham S. Meijnikman, Max Nieuwdorp, and Bernd Schnabl. Endogenous ethanol production in health and disease. *Nature Reviews Gastroenterology & Hepatology*, 21, 08 2024. URL: https://www.nature.com/articles/s41575-024-00937-w, doi:10.1038/s41575-024-00937-w.

[156] Chiara Milano, Domeniko Hoxhaj, Marta Del Chicca, Alessia Pascazio, Davide Paoli, Luca Tommasini, Andrea Vergallo, Chiara Pizzanelli, Gloria Tognoni, Angelo Nuti, Roberto Ceravolo, Gabriele Siciliano, Harald Hampel, and Filippo Baldacci. Alzheimer's disease and neurosyphilis: Meaningful commonalities and differences of clinical phenotype and pathophysiological biomarkers. *Journal of Alzheimer's Disease*, 94, 07 2023. doi:10.3233/jad-230170.

[157] Artem Minalyan, Lilit Gabrielyan, David Scott, Jonathan Jacobs, and Joseph R. Pisegna. The gastric and intestinal microbiome: Role of proton pump inhibitors. *Current Gastroenterology Reports*, 19(8):42, Jul 2017. doi:10.1007/s11894-017-0577-6.

[158] Maria B. Misiura, Brittany Butts, Bruno Hammerschlag, Chinkuli Munkombwe, Arianna Bird, Mercedes Fyffe, Asia Hemphill, Vonetta M. Dotson, and Whitney Wharton. Intersectionality in alzheimer's disease: The role of female sex and black american race in the development and prevalence of alzheimer's disease. *Neurotherapeutics*, 20(4):1019–1036, 2023. URL: https://www.sciencedirect.com/science/article/pii/S1878747923019098, doi:10.1007/s13311-023-01408-x.

[159] Sabrina Mitchell, Teresa Welch-Burke, Logan Dumitrescu, Jefferson P Lomenick, Deborah G Murdock, Dana C Crawford, and Marshall Summar. Peptide tyrosine tyrosine levels are increased in patients with urea cycle disorders. *Molecular genetics and metabolism*, pages 39–42, 02 2012. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC3336020/, doi:10.1016/j.ymgme.2012.02.011.

[160] Mariya Timotey Miteva, Davide Laurenti, Roberto Mattioli, Daniel Di Risola, Alessia Mariano, and Luciana Mosca. Vitamin deficiencies and alzheimer's disease: evidence and implications for supplementation. *Frontiers in nutrition*, page 1676497, 02 2026. URL: https://pubmed.ncbi.nlm.nih.gov/41769656/, doi:10.3389/fnut.2026.1676497.

[161] Maria Mostyka, Jinru Shia, William L. Neumann, Christa L. Whitney-Miller, Michael Feely, and Rhonda K. Yantiss. Clinicopathologic features of varicella zoster virus infection of the upper gastrointestinal tract. *American Journal of Surgical Pathology*, 45, 2021. doi:10.1097/pas.0000000000001576.

[162] Amritpal Mudher and Simon Lovestone. Alzheimer's disease - do tauists and baptists finally shake hands? *Trends in Neurosciences*, 25, 01 2002. doi:10.1016/s0166-2236(00)02031-2.

[163] Caitlin C. Murdoch and Eric P. Skaar. Nutritional immunity: the battle for nutrient metals at the host-pathogen interface. *Nature Reviews Microbiology*, 20, 11 2022. URL: https://www.nature.com/articles/s41579-022-00745-6, doi:10.1038/s41579-022-00745-6.

[164] Thomaz Funari Neto and Emanuela Ferreira Esper. "ozempic face": Aesthetic consequence, unintended muscle loss, or a warning sign of maladaptive weight reduction? *International Health Sciences Review*, 1, 09 2025. doi:10.70164/ihsr.v1i4.49.

[165] Roch A. Nianogo, Amy Rosenwohl-Mack, Kristine Yaffe, Anna Carrasco, Coles M. Hoffmann, and Deborah E. Barnes. Risk factors associated with alzheimer disease and related dementias by sex and race and ethnicity in the us. *JAMA Neurology*, 79, 06 2022. doi:10.1001/jamaneurol.2022.0976.

[166] John M. Nolan, Rebecca Power, Alan N. Howard, Paula Bergin, Warren Roche, Alfonso Prado-Cabrero, George Pope, John Cooke, Tommy Power, and Ríona Mulcahy. Supplementation with carotenoids, omega-3 fatty acids, and vitamin e has a positive effect on the symptoms and progression of alzheimer's disease. *Journal of Alzheimer's Disease*, 90, 10 2022. doi:10.3233/jad-220556.

[167] Adrian L Oblak, Peter B Lin, Kevin P Kotredes, Ravi S Pandey, Dylan Garceau, Harriet M Williams, Asli Uyar, Rita O'Rourke, Sarah O'Rourke, Cynthia Ingraham, Daria Bednarczyk, Melisa Belanger, Zackary A Cope, Gabriela J Little, Sean-Paul G Williams, Carl Ash, Adam Bleckert, Tim Ragan, Benjamin A Logsdon, Lara M Mangravite, Stacey J Sukoff Rizzo, Paul R Territo, Gregory W Carter, Gareth R Howell, Michael Sasner, and Bruce T Lamb. Comprehensive evaluation of the 5xfad mouse model for preclinical testing applications: A model-ad study. *Frontiers in aging neuroscience*, page 713726, 07 2021. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC8346252/, doi:10.3389/fnagi.2021.713726.

[168] Ashley O'Mara, Bhakti Pradip Mody, Marco Mammi, Thomas Simjian, Kyrellos Ghattas, Srilekha Kaliki, Ngoc Mai Le, Aaron Liew, Mattia Migliore, and Rania A. Mekary. The effect of glp-1 receptor agonists on cognition in nondiabetic patients with mild cognitive impairment or alzheimer's disease: a meta-analysis of randomized controlled trials. *Neurological Sciences*, 47(1):149, Jan 2026. doi:10.1007/s10072-025-08602-z.

[169] James Owusu-Kwarteng, Dominic Agyei, Fortune Akabanda, Richard Atinpoore Atuna, and Francis Kweku Amagloh. Plant-based alkaline fermented foods as sustainable sources of nutrients and health-promoting bioactive compounds. *Frontiers in Sustainable Food Systems*, 6, 06 2022. doi:10.3389/fsufs.2022.885328.

[170] Francesco Panza, Vittorio Dibello, Rodolfo Sardone, Roberta Zupo, Fabio Castellana, Ivana Leccisotti, Maria Claudia Moretti, Mario Altamura, Antonello Bellomo, Antonio Daniele, Vincenzo Solfrizzi, Emanuela Resta, and Madia Lozupone. Successes and failures: the latest advances in the clinical development of amyloid-upbeta-targeting monoclonal antibodies for treating alzheimer's disease. *Expert Opinion on Biological Therapy*, 25, 03 2025. doi:10.1080/14712598.2025.2463963.

[171] Ah-Mee Park and Ikuo Tsunoda. Helicobacter pylori infection in the stomach induces neuroinflammation: the potential roles of bacterial outer membrane vesicles in an animal model of alzheimer's disease. *Inflammation and Regeneration*, 42(1):39, Sep 2022. doi:10.1186/s41232-022-00224-8.

[172] Ligi Paul. Diet, nutrition and telomere length. *The Journal of Nutritional Biochemistry*, 22(10):895–901, 2011. URL: https://www.sciencedirect.com/science/article/pii/S0955286311000052, doi:10.1016/j.jnutbio.2010.12.001.

[173] Tzu-Rong Peng, Hung-Hong Lin, Tzu-Ling Tseng, and Ta-Wei Wu. Effectiveness of probiotic supplements on cognitive function in mild cognitive impairment and alzheimer's disease: A meta-analysis of randomized controlled trials. *General Hospital Psychiatry*, 97:284–293, 2025. URL: https://www.sciencedirect.com/science/article/pii/S0163834325002142, doi:10.1016/j.genhosppsych.2025.11.004.

[174] Jacob M. Pfaffinger, Kallie E. Hays, Jason Seeley, Priyadharshine Ramesh Babu, and Rebecca Ryznar. Gut dysbiosis as a potential driver of parkinson's and alzheimer's disease pathogenesis. *Frontiers in Neuroscience*, 19, 08 2025. doi:10.3389/fnins.2025.1600148.

[175] Thomas Piekut, MikoUx0142[bad char vv=322]aj HurUx0142[bad char vv=322]a, Natalia Banaszek, Paulina Szejn, Jolanta Dorszewska, Wojciech Kozubski, and MichaUx0142[bad char vv=322] Prendecki. Infectious agents and alzheimer's disease. *Journal of Integrative Neuroscience*, 21, 2022. doi:10.31083/j.jin2102073.

[176] Burkhard Poeggeler, Sandeep Kumar Singh, and Miguel A. Pappolla. Tryptophan in nutrition and health. *International Journal of Molecular Sciences*, 23, 05 2022. doi:10.3390/ijms23105455.

[177] Baruh Polis and Abraham O. Samson. Addressing the discrepancies between animal models and human alzheimer's disease pathology: Implications for translational research. *Journal of Alzheimer's Disease*, 98, 04 2024. doi:10.3233/jad-240058.

[178] Elena Porras-García, Irene Fernández-Espada Calderón, Juan Gavala-González, and José Carlos Fernández-García. Potential neuroprotective effects of fermented foods and beverages in old age: a systematic review. *Frontiers in Nutrition*, 10, 06 2023. doi:10.3389/fnut.2023.1170841.

[179] Nelsan Pourhadi, Janet Janbek, Christina JensenDahm, Christiane Gasse, Thomas Munk Laursen, and Gunhild Waldemar. Proton pump inhibitors and dementia: A nationwide populationbased study. *Alzheimer's & Dementia*, 20, 02 2024. doi:10.1002/alz.13477.

[180] Joseph R. Prohaska and Bruce Brokate. Dietary copper deficiency alters protein levels of rat dopamine upbeta-monooxygenase and tyrosine monooxygenase. *Experimental Biology and Medicine*, 226, 03 2001. doi:10.1177/153537020122600307.

[181] Deepti Putcha, Yuta Katsumi, Alexandra Touroutoglou, Ani Eloyan, Alexander Taurone, Maryanne Thangarajah, Paul Aisen, Jeffrey L. Dage, Tatiana Foroud, Clifford R. Jack, Joel H. Kramer, Kelly H. Nudelman, Rema Raman, Prashanthi Vemuri, Alireza Atri, Gregory S. Day, Ranjan Duara, Neill R. Graff-Radford, Ian M. Grant, Lawrence S. Honig, Erik B. Johnson, David T. Jones, Joseph C. Masdeu, Mario F. Mendez, Erik Musiek, Chiadi U. Onyike, Meghan Riddle, Emily Rogalski, Stephen Salloway, Sharon Sha, R. Scott Turner, Thomas S. Wingo, David A. Wolk, Kyle Womack, Maria C. Carrillo, Gil D. Rabinovici, Bradford C. Dickerson, Liana G. Apostolova, Dustin B. Hammers, the LEADS Consortium, Laurel Beckett, Bernardino Ghetti, Lea T. Grinberg, Dustin Hammers, Clifford R. Jack, Kala Kirby, Robert Koeppe, Walter A. Kukull, Renaud La Joie, Julien Lagarde, Melissa E. Murray, Kathy Newell, Angelina Polsinelli, Arthur Toga, David Clark, Ian Grant, and Lawrence S. Honig. Heterogeneous clinical phenotypes of sporadic early-onset alzheimer's disease: a neuropsychological data-driven approach. *Alzheimer's Research & Therapy*, 17(1):38, Feb 2025. doi:10.1186/s13195-025-01689-8.

[182] Lili Qiu, Qianqian Huang, Wenhao Li, Qian Zhang, Jun Zhou, Juan Chen, Yixuan Li, Ran Wang, Pengjie Wang, Siyuan Liu, Bing Fang, and Xiaoyu Wang. Aging influences protein digestion, absorption and amino acid metabolism. *Biogerontology*, 26(4):146, Jul 2025. doi:10.1007/s10522-025-10289-w.

[183] Zehui Qiu, Yuyao Shi, Yi Wu, Yao Zheng, Wenzheng Shi, Mingyu Yin, Long Zhang, and Xichang Wang. Altered digestive and absorptive characteristics of food proteins in older adults: Mechanisms, model applications, and innovative strategies. *Comprehensive Reviews in Food Science and Food Safety*, 25, 01 2026. doi:10.1111/1541-4337.70364.

[184] Khanh vinh quUx1ed1[bad char vv=7889]c Lu'o'ng and Lan Thi Hoàng NguyUx1ec5[bad char vv=7877]n. Role of thiamine in alzheimer's disease. *American Journal of Alzheimer's Disease & Other Dementias®*, 26, 12 2011. doi:10.1177/1533317511432736.

[185] Aneela Rahman, Hande Jackson, Hollie Hristov, Richard S. Isaacson, Nabeel Saif, Teena Shetty, Orli Etingin, Claire Henchcliffe, Roberta Diaz Brinton, and Lisa Mosconi. Sex and gender driven modifiers of alzheimer's: The role for estrogenic control across age, race, medical, and lifestyle risks. *Frontiers in Aging Neuroscience*, 11, 11 2019. doi:10.3389/fnagi.2019.00315.

[186] Satish K. Raut, Kulwinder Singh, Shridhar Sanghvi, Veronica Loyo-Celis, Liyah Varghese, Ekam R. Singh, Shubha Gururaja Rao, and Harpreet Singh. Chloride ions in health and disease. *Bioscience Reports*, 44, 05 2024. doi:10.1042/bsr20240029.

[187] May J Reed, Mamatha Damodarasamy, and William A Banks. The extracellular matrix of the blood-brain barrier: structural and functional roles in health, aging, and alzheimer's disease. *Tissue barriers*, page 1651157, 09 2019. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC6866683/, doi:10.1080/21688370.2019.1651157.

[188] Kryssia Isabel Rodriguez-Castro, Marilisa Franceschi, Antonino Noto, Chiara Miraglia, Antonio Nouvenne, Gioacchino Leandro, Tiziana Meschi, Gian Luigi De' Angelis, and Francesco Di Mario. Clinical manifestations of chronic atrophic gastritis. *Acta bio-medica : Atenei Parmensis*, pages 88–92, 12 2018. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC6502219/pdf/ACTA-89-88.pdf, doi:10.23750/abm.v89i8-S.7921.

[189] Gustavo C. Román, Oscar Mancera-Páez, and Camilo Bernal. Epigenetic factors in late-onset alzheimer's disease: Mthfr and cth gene polymorphisms, metabolic transsulfuration and methylation pathways, and b vitamins. *International Journal*

of Molecular Sciences, 20, 01 2019. doi:10.3390/ijms20020319.

[190] Eric L. Ross, Marc S. Weinberg, and Steven E. Arnold. Cost-effectiveness of aducanumab and donanemab for early alzheimer disease in the us. *JAMA Neurology*, 79, 05 2022. doi:10.1001/jamaneurol.2022.0315.

[191] Lina M. Ruiz, Erik L. Jensen, Yancing Rossel, German I. Puas, Alvaro M. Gonzalez-Ibanez, Rodrigo I. Bustos, David A. Ferrick, and Alvaro A. Elorza. Non-cytotoxic copper overload boosts mitochondrial energy metabolism to modulate cell proliferation and differentiation in the human erythroleukemic cell line k562. *Mitochondrion*, 29:18–30, 2016. URL: https://www.sciencedirect.com/science/article/pii/S1567724916300241, doi:10.1016/j.mito.2016.04.005.

[192] RM Russell. Changes in gastrointestinal function attributed to aging. *The American Journal of Clinical Nutrition*, 55(6):1203S–1207S, 1992. URL: https://www.sciencedirect.com/science/article/pii/S0002916523317222, doi:10.1093/ajcn/55.6.1203S.

[193] Tanya L. Russell, Rosemary R. Berardi, Jeffrey L. Barnett, Lambros C. Dermentzoglou, Kathleen M. Jarvenpaa, Stephen P. Schmaltz, and Jennifer B. Dressman. Upper gastrointestinal ph in seventy-nine healthy, elderly, north american men and women. *Pharmaceutical Research*, 10(2):187–196, Feb 1993. doi:10.1023/A:1018970323716.

[194] Arisha Salam and Mohammed Salman Hyder. Semaglutide and the skin: A brief review of dermatologic implications. *Cosmoderma*, 5, 08 2025. doi:10.25259/csdm_97_2025.

[195] Crystal Sang, Sasha A. Philbert, Danielle Hartland, Richard. D Unwin, Andrew W. Dowsey, Jingshu Xu, and Garth J. S. Cooper. Coenzyme a-dependent tricarboxylic acid cycle enzymes are decreased in alzheimer's disease consistent with cerebral pantothenate deficiency. *Frontiers in Aging Neuroscience*, 14, 06 2022. doi:10.3389/fnagi.2022.893159.

[196] Genevieve Saw, Ling-Xiao Yi, Eng King Tan, and Zhi Dong Zhou. Evidence suggesting that alzheimer's disease may be a transmissible disorder. *International Journal of Molecular Sciences*, 26, 01 2025. doi:10.3390/ijms26020508.

[197] Melissa Scholefield, Stephanie J. Church, Jingshu Xu, Stefano Patassini, and Garth J.S. Cooper. Localized pantothenic acid (vitamin b5) reductions present throughout the dementia with lewy bodies brain. *Journal of Parkinson's Disease*, 14, 07 2024. doi:10.3233/jpd-240075.

[198] Merve Hazal Ser, Fatma Zehra ÇalkuUx015f[bad char vv=351]u, Nursena Erener, Orhan DestanoUx011f[bad char vv=287]lu, ErtuUx011f[bad char vv=287]rul Kykm, and Aksel Siva. Auto brewery syndrome from the perspective of the neurologist. *Journal of Forensic and Legal Medicine*, 96:102514, 2023. URL: https://www.sciencedirect.com/science/article/pii/S1752928X2300032X, doi:10.1016/j.jflm.2023.102514.

[199] Harsh Shah, Fereshteh Dehghani, Marjan Ramezan, Ritchel B. Gannaban, Zobayda Farzana Haque, Fatemeh Rahimi, Soheil Abbasi, and Andrew C. Shin. Revisiting the role of vitamins and minerals in alzheimer's disease. *Antioxidants*, 12, 02 2023. doi:10.3390/antiox12020415.

[200] Oliver M. Shannon, John C. Mathers, Emma Stevenson, and Mario Siervo. Healthy dietary patterns, cognition and dementia risk: current evidence and context. *Proceedings of the Nutrition Society*, 05 2025. doi:10.1017/s0029665125100050.

[201] Xiangli Shao, Zhenglin Yang, Zhihao Xu, Weijie Guo, Whitney Lewis, Boyi Gan, and Yi Lu. Simultaneous imaging of cu $^{+}$ and cu $^{2+}$ in neural cells using dnazyme probes reveals mechanistic link between copper redox imbalance and amyloid pathology. *Angewandte Chemie International Edition*, 02 2026. doi:10.1002/anie.202524237.

[202] Nitin Shivappa, Michael D. Wirth, Thomas G. Hurley, and James R. Hébert. Association between the dietary inflammatory index (dii) and telomere length and creactive protein from the national health and nutrition examination survey1999-2002. *Molecular Nutrition & Food Research*, 61, 04 2017. doi:10.1002/mnfr.201600630.

[203] Cara Lynn Shultz. People on glp-1s are getting an old-timey sailorsđisease. *PEOPLE*, 02 2026. URL: https://people.com/glp-1s-linked-to-old-timey-sailors-disease-scurvy-11905053.

[204] Manjinder Singh, Maninder Kaur, Hitesh Kukreja, Rajan Chugh, Om Silakari, and Dhandeep Singh. Acetylcholinesterase inhibitors as alzheimer therapy: From nerve toxins to neuroprotection. *European Journal of Medicinal Chemistry*, 70:165–188, 2013. URL: https://www.sciencedirect.com/science/article/pii/S0223523413006314, doi:10.1016/j.ejmech.2013.09.050.

[205] Dion Smink, Alberto Juan, Boelo Schuur, and Sascha A. Kersten. Understanding the role of choline chloride in deep eutectic solvents used for biomass delignification. *Industrial & Engineering Chemistry Research*, 58(36):16348–16357, 2019. arXiv:https://doi.org/10.1021/acs.iecr.9b03588, doi:10.1021/acs.iecr.9b03588.

[206] Jiangman Song, Lu Yang, Di Nan, Qihua He, You Wan, and Huailian Guo. Histidine alleviates impairments induced by chronic cerebral hypoperfusion in mice. *Frontiers in Physiology*, 9, 06 2018. doi:10.3389/fphys.2018.00662.

[207] Jyoti Srivastava and Sanjay Premi. Melanin as a redox hub: Linking energy flow to genome stability and immune control. *Antioxidants & redox signaling*, page 15230864261421542, 02 2026. URL: https://pubmed.ncbi.nlm.nih.gov/41680087/, doi:10.1177/15230864261421542.

[208] Renee Stamation. Endogenous ethanol production in the human alimentary tract: A literature review. *Journal of Gastroenterology and Hepatology*, 40, 04 2025. doi:10.1111/jgh.16869.

[209] Omme Fatema Sultana, Raksa Andalib Hia, and P. Hemachandra Reddy. A combinational therapy for preventing and delaying the onset of alzheimer's disease: A focus on probiotic and vitamin co-supplementation. *Antioxidants*, 13, 02 2024. doi:10.3390/antiox13020202.

[210] Masako Suzuki, Tao Wang, Diana Garretto, Carmen R. Isasi, Wellington V. Cardoso, John M. Greally, and Loredana Quadro. Disproportionate vitamin a deficiency in women of specific ethnicities linked to differences in allele frequencies of vitamin a-related polymorphisms. *Nutrients*, 13, 06 2021. doi:10.3390/nu13061743.

[211] Laszlo Sztriha and A. Lorris Betz. Oleic acid reversibly opens the blood-brain barrier. *Brain Research*, 550, 06 1991. doi:10.1016/0006-8993(91)91326-v.

[212] Leon M Tai, Deebika Balu, Evangelina Avila-Munoz, Laila Abdullah, Riya Thomas, Nicole Collins, Ana Carolina Valencia-Olvera, and Mary Jo LaDu. Efad transgenic mice as a human apoe relevant preclinical model of alzheimer's disease.

Journal of lipid research, pages 1733–1755, 04 2017. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC5580905/, doi:10.1194/jlr.R076315.

[213] Michihiro Takada, Shin Tanaka, Koh Tanaka, Tamao Tsukie, Masako TsukamotoYasui, Katsuya Suzuki, Yasushi Noguchi, Akira Imaizumi, Makoto Ishii, and Takeshi Ikeuchi. Effects of an essential amino acid mixture on behavioral and psychological symptoms of dementia and executive function in patients with alzheimer's disease: A doubleblind, randomized, placebocontrolled exploratory clinical trial. *International Journal of Geriatric Psychiatry*, 37, 09 2022. doi:10.1002/gps.5782.

[214] Jian-Ping Tang and Srikumaran Melethil. Effect of aging on the kinetics of blood–brain barrier uptake of tryptophan in rats. *Pharmaceutical Research*, 12(7):1085–1091, Jul 1995. doi:10.1023/A:1016283003747.

[215] NeuroLaunch editorial team. Semaglutide mental side effects: Navigating the psychological impact of weight loss medication. 02 2025. URL: https://neurolaunch.com/semaglutide-mental-side-effects/.

[216] Andrew F. Teich, Katarnut Tobunluepop, Juan C. Troncoso, ShihHsiu Wang, Zihan Wang, Benjamin Wolozin, Jesse Mez, Lindsay A. Farrer, Mark W. Logue, Adam Labadorf, Nicholas K. O'Neill, Dennis W. Dickson, Brittany N. Dugger, Margaret E. Flanagan, Matthew P. Frosch, Marla Gearing, LeeWay Jin, Julia Kofler, Richard Mayeux, Ann McKee, Carol A. Miller, Melissa E. Murray, Peter T. Nelson, Richard J. Perrin, Julie A. Schneider, and Thor D. Stein. Novel differentially expressed genes and multiple biological pathways for alzheimer's disease identified in brain tissue from african american donors. *Alzheimer's & Dementia*, 21, 10 2025. doi:10.1002/alz.70629.

[217] Torsak Tippairote, Pruettithada Hoonkaew, Aunchisa Suksawang, and Prayfan Tippairote. Glucagon suppression under glp-1ra therapy: hidden trade-offs for muscle and resilience. *Diabetology International*, 17(1):2, Nov 2025. doi:10.1007/s13340-025-00856-4.

[218] Jorge Urbina, Luis Eduardo SalinasRuiz, César Valenciano, and Benjamin Clapp. Micronutrient and nutritional deficiencies associated with <scp>glp</scp> 1 receptor agonist therapy: A narrative review. *Clinical Obesity*, 16, 02 2026. doi:10.1111/cob.70070.

[219] Tanmaykumar Varma, Pradnya Kamble, Madhavi Kumari, Vineet Diwakar, and Prabha Garg. *Vitamin-Based Derivatives for the Management of Alzheimer's Disease*, pages 317–344. Springer Nature Singapore, Singapore, 2023. doi:10.1007/978-981-99-6038-5_12.

[220] Ramon Velazquez, Eric Ferreira, Sara Knowles, Chaya Fux, Alexis Rodin, Wendy Winslow, and Salvatore Oddo. Lifelong choline supplementation ameliorates alzheimer's disease pathology and associated cognitive deficits by attenuating microglia activation. *Aging Cell*, 18, 12 2019. doi:10.1111/acel.13037.

[221] Ramon Velazquez, Wendy Winslow, and Marc A Mifflin. Choline as a prevention for alzheimer's disease., 02 2020. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC7041773/, doi:10.18632/aging.102849.

[222] Natascha Verhagen, Andy Wiranata Wijaya, Attila Teleki, Muhammad Fadhlullah, Andreas Unsld, Martin Schilling, Christoph Heinrich, and Ralf Takors. Comparison of l-tyrosine containing dipeptides reveals maximum atp availability for l-prolyl-l-tyrosine in cho cells. *Engineering in life sciences*, pages 384–394, 06 2020. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC7481768/, doi:10.1002/elsc.202000017.

[223] Xiaodi Wang, Wen Chen, Jie Zhang, Aiman Khan, Liangpeng Li, Fuhua Huang, Zhibing Qiu, Liming Wang, and Xin Chen. Critical role of adamts2 (a disintegrin and metalloproteinase with thrombospondin motifs 2) in cardiac hypertrophy induced by pressure overload. *Hypertension*, 69(6):1060–1069, 2017. URL: https://www.ahajournals.org/doi/abs/10.1161/HYPERTENSIONAHA.116.08581, arXiv:https://www.ahajournals.org/doi/pdf/10.1161/HYPERTENSIONAHA.116.08581, doi:10.1161/HYPERTENSIONAHA.116.08581.

[224] Yuting Wang, Xiaoxia Lin, Lan Cheng, Xinxin Cheng, Jianyun He, and Shufang Xia. Dietary patterns and risk of dementia in elderly individuals: A systematic review and meta-analysis of cohort studies. *Nutrition Reviews*, page nuaf121, 07 2025. arXiv:https://academic.oup.com/nutritionreviews/advance-article-pdf/doi/10.1093/nutrit/nuaf121/63730253/nuaf121.pdf, doi:10.1093/nutrit/nuaf121.

[225] Andrew E. Welchman and Zoe Kourtzi. Solving the 'goldilocks problem' in dementia clinical trials with multimodal ai. *The Journal of Prevention of Alzheimer's Disease*, 13(1):100397, 2026. URL: https://www.sciencedirect.com/science/article/pii/S2274580725003395, doi:10.1016/j.tjpad.2025.100397.

[226] Maodong Wu, Lijuan Gao, Qinglun Su, Haibing Wang, Qin Zhao, and Hongxing Wang. Global burden, spatiotemporal disparities, and sex-specific trends of early- and late-onset alzheimer's disease and other dementias: a comparative analysis and projections to 2035. *BMC Neurology*, 26(1):1, Nov 2025. doi:10.1186/s12883-025-04530-9.

[227] Peng Wu, Shao-Xiu Pu, Mao-Shuai Jiang, Ze-Jun Hu, Chang-Yong Li, Didier Dupont, and Xiao-Dong Chen. Enhancing food digestion and nutrient absorption through pre-digestion: Tailored strategies for special populations. *Food & Medicine Homology*, 2, 12 2025. doi:10.26599/fmh.2025.9420069.

[228] Peng Wu, Ping Zhang, and Xiao Dong Chen. Chapter five - assessing food digestion in the elderly using in vitro gastrointestinal models. volume 114 of *Advances in Food and Nutrition Research*, pages 273–300. Academic Press, 2025. URL: https://www.sciencedirect.com/science/article/pii/S1043452624000986, doi:10.1016/bs.afnr.2024.09.004.

[229] Jingshu Xu, Stephanie J Church, Stefano Patassini, Paul Begley, Henry J Waldvogel, Maurice A Curtis, Richard M Faull, Richard D Unwin, and Garth S Cooper. Evidence for widespread, severe brain copper deficiency in alzheimer's dementiadagger. *Metallomics*, 9(8):1106–1119, 06 2017. arXiv:https://academic.oup.com/metallomics/article-pdf/9/8/1106/34732504/c7mt00074j.pdf, doi:10.1039/c7mt00074j.

[230] Xinming Xu, Guliyeerke Jigeer, David Andrew Gunn, Yizhou Liu, Xinrui Chen, Yi Guo, Yaqi Li, Xuelan Gu, Yanyun Ma, Jiucun Wang, Sijia Wang, Liang Sun, Xu Lin, and Xiang Gao. Facial aging, cognitive impairment, and dementia risk. *Alzheimer's Research & Therapy*, 16(1):245, Nov 2024. doi:10.1186/s13195-024-01611-8.

[231] Liya Yan, Bo. Zhou, Shailja Nigdikar, Xiaohong Wang, Janet Bennett, and Ann Prentice. Effect of apolipoprotein e genotype on vitamin k status in healthy older adults from china and the uk. *British Journal of Nutrition*, 94, 12 2005.

doi:10.1079/bjn20051578.

[232] Meifeng Yang, Junye Miao, Joshua Rizak, Rongwei Zhai, Zhengbo Wang, Tanzeel Huma, Ting Li, Na Zheng, Shihao Wu, Yingwei Zheng, Xiaona Fan, Jianzhen Yang, Jianhong Wang, Shangchuan Yang, Yuanye Ma, Longbao Lü, Rongqiao He, and Xintian Hu. Alzheimer's disease and methanol toxicity (part 2): Lessons from four rhesus macaques (*macaca mulatta*) chronically fed methanol. *Journal of Alzheimer's Disease*, 41, 07 2014. doi:10.3233/jad-131532.

[233] Lian Ye, Rui Liang, Xiaolei Liu, Jun Li, Jirong Yue, and Xinjun Zhang. Frailty and sarcopenia: A bibliometric analysis of their association and potential targets for intervention. *Ageing Research Reviews*, 92:102111, 2023. URL: https://www.sciencedirect.com/science/article/pii/S1568163723002702, doi:10.1016/j.arr.2023.102111.

[234] Awgichew Shewasinad Yehualashet, Ermiyas Endewunet Melaku, Demissew Shenkute Gebreyes, Tilahun Alelign Wassie, and Berhanu Yitayew Sahilu. The bilateral cross communication in microbiota-gut-brain axis as a promising therapeutic target for alzheimer's disease: a focus on neuroinflammation. *Discover Medicine*, 2(1):33, Feb 2025. doi:10.1007/s44337-025-00220-0.

[235] Masaru Yonekawa, Tomohiro Okabe, Yasumasa Asamoto, and Michiya Ohta. A case of hereditary ceruloplasmin deficiency with iron deposition in the brain associated with chorea, dementia, diabetes mellitus and retinal pigmentation: Administration of fresh-frozen human plasma. *European Neurology*, 42, 10 1999. doi:10.1159/000008091.

[236] Kyung Man You, Claire-Lise Rosenfield, and Douglas C. Knipple. Ethanol tolerance in the yeast*saccharomyces cerevisiae*is dependent on cellular oleic acid content. *Applied and Environmental Microbiology*, 69, 03 2003. doi:10.1128/aem.69.3.1499-1503.2003.

[237] Jingpu Zhang, Yanlei Liu, Xiao Zhi, Li Xu, Jie Tao, Daxiang Cui, and Tie Fu Liu. Tryptophan catabolism via the kynurenine pathway regulates infection and inflammation: from mechanisms to biomarkers and therapies. *Inflammation Research*, 73(6):979–996, Jun 2024. doi:10.1007/s00011-024-01878-5.

[238] Junhan Zhang, Celi Yang, Yuefeng Tan, Muzi Na, Evaniya Shakya, Da Zhang, Xiaojin Shi, Xiaona Na, Zhihui Li, John S Ji, Yucheng Yang, and Ai Zhao. Region-specific brain structural modulation and amyloid-beta pathology associated with dietary biotin: insights into dementia neuropathology. *EBioMedicine*, page 106155, 02 2026. URL: https://pubmed.ncbi.nlm.nih.gov/41679193/, doi:10.1016/j.ebiom.2026.106155.

[239] Shu Zhang, Rei Otsuka, Yukiko Nishita, Chikako Tange, Makiko Tomida, Fujiko Ando, Hiroshi Shimokata, and Hidenori Arai. Twenty-year prospective cohort study of the association between a japanese dietary pattern and incident dementia: the nils-lsa project. *European Journal of Nutrition*, 62(4):1719–1729, Jun 2023. doi:10.1007/s00394-023-03107-x.

[240] Sirui Zhang, Yi Xiao, Yangfan Cheng, Yuanzheng Ma, Jiyong Liu, Chunyu Li, and Huifang Shang. Associations of sugar intake, high-sugar dietary pattern, and the risk of dementia: a prospective cohort study of 210,832 participants. *BMC Medicine*, 22(1):298, Jul 2024. doi:10.1186/s12916-024-03525-6.

[241] Han Zhao, Shiyao Wang, and Xu Li. Dna damage accumulation in aging brain and its links to alzheimer's disease progression. *Genome Instability & Disease*, 3(3):172–178, Jun 2022. doi:10.1007/s42764-022-00069-y.

[242] Yuhai Zhao, Vivian Jaber, and Walter J. Lukiw. Secretory products of the human gi tract microbiome and their potential impact on alzheimer's disease (ad): Detection of lipopolysaccharide (lps) in ad hippocampus. *Frontiers in Cellular and Infection Microbiology*, 7, 07 2017. doi:10.3389/fcimb.2017.00318.

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2. Free software including Linux, R, LaTex etc.

3. Thanks everyone who contributed incidental support.

Appendix A: Statement of Conflicts

No specific funding was used in this effort and there are no relationships with others that could create a conflict of interest. I would like to develop these ideas further and have obvious bias towards making them appear successful. Barbara Cade, the dog owner, has worked in the pet food industry but this does not likely create a conflict. We have no interest in the makers of any of the products named in this work.

Appendix B: About the Authors and Facility

This work was performed at a dog rescue run by Barbara Cade and housed in rural Georgia. The author of this report ,Mike Marchywka, has a background in electrical engineering and has done extensive research using free online literature sources. I hope to find additional people interested in critically examining the results and verify that they can be reproduced effectively to treat other dogs.

Appendix C: Symbols, Abbreviations and Colloquialisms

TERM	definition and meaning

Appendix D: General caveats and disclaimer

This document was created in the hope it will be interesting to someone including me by providing information about some topic that may include personal experience or a literature review or description of a speculative theory or idea. There is no assurance that the content of this work will be useful for any paricular purpose.

All statements in this document were true to the best of my knowledge at the time they were made and every attempt is made to assure they are not misleading or confusing. However, information provided by others and observations that can be manipulated by unknown causes (”gaslighting”) may be misleading. Any use of this information should be preceded by validation including replication where feasible. Errors may enter into the final work at every step from conception and research to final editing.

Documents labelled ”NOTES” or ”not public” contain substantial informal or speculative content that may be terse and poorly edited or even sarcastic or profane. Documents labelled as ”public” have generally been edited to be more coherent but probably have not been reviewed or proof read.

Generally non-public documents are labelled as such to avoid confusion and embarassment and should be read with that understanding.

Appendix E: Citing this as a tech report or white paper

Note: This is mostly manually entered and not assured to be error free.
This is tech report MJM-2026-001.

Version	Date	Comments
0.01	2026-01-17	Create from empty.tex template
-	March 18, 2026	version 0.00 MJM-2026-001
1.0	20xx-xx-xx	First revision for distribution

Released versions,
build script needs to include empty releases.tex

Version	Date	URL

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Supporting files. Note that some dates,sizes, and md5's will change as this is rebuilt.

This really needs to include the data analysis code but right now it is auto generated picking up things from prior build in many cases

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1538 Aug 14 2021 /home/documents/latex/share/includes/recent_template.tex 49763d2c29f74e4b54fa53b25c2cc439
7217 May 15 2023 /home/documents/latex/share/overrides/mol2chemfig.sty 5aa81c197d462c2153a794ccee9881ef
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33495 Jan 30 2023 /usr/share/texlive/texmf-dist/bibtex/bst/urlbst/plainurl.bst 46
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